DIRECTED MOVEMENTS AND SELECTIVE ADHESION OF EMBRYONIC AMPHIBIAN CELLS [1]

PHILIP L. TOWNES [2] AND JOHANNES HOLTFRETER
Departments of Biology and Anatomy, The University of Rochester, Rochester, New York

TWENTY-SEVEN FIGURES

INTRODUCTION

One of the most striking features of early vertebrate development is the transformation of a spherical egg into a body of about equal size in which groups of cells have shifted into specific arrangements which foreshadow the tissue pattern of the adult organism. This transition occurs principally between the blastula and neurula stages. Throughout the history of embryology, these shiftings of cell associations and their subsequent segregation into germ layers and tissues have been of foremost interest to students of embryogenesis.

It was these transformations which prompted His (1874) to propose his famous, though no longer acceptable, interpretation of embryological phenomena in mechanistic terms. Roux (1894, 1896), in an attempt to relate tissue formation to the kinetic properties of individual cells, teased early amphibian embryos apart and recorded the movements and re-aggregations of the free cells in a medium of diluted egg white. Roux claimed to have found evidence that the cells produce diffusible substances which either attract or repel other cells in the immediate vicinity and which he considered

[1] Revision of a thesis submitted in partial fulfillment of the requirements for the degree of Doctor of Philosophy, Department of Biology, The University of Rochester.

[2] Predoctoral Research Fellow, National Cancer Institute, U. S. Public Health Service 1951–1952. Present Address: Department of Anatomy, School of Medicine, University of Rochester.

to be essential agents for bringing about normal tissue aggregations and segregations. These claims about cytotropisms remained unchallenged for many years. When, finally, Voigtländer ('32) and Kuhl ('37) subjected them to a careful reinvestigation, they could not be confirmed. These workers found that no attraction nor repulsion between the separated cells could be detected and that the path of the individual motile cells is entirely at random; but when two or more cells meet by chance, they tend to unite, even if they are derived from different species or germ layers. Kuhl concluded that the phenomena were largely "pathological" and as such unsuited to serve as a basis for interpreting normal histogenesis.

A more optimistic attitude was adopted by Holtfreter ('38, '39, '43a, b, c, '44, '47a, b, '48) who found that isolation of embryonic amphibian cells in an appropriate medium need not in itself interfere with their viability or differentiation, and that the method of isolating embryonic tissues or their cellular constituents, combined with the method of reuniting them in new combinations, offers valuable opportunities for a study of the factors involved in organogenesis. From his studies we may extract the following points relevant to our discussion.

1. The cellular mass movements of gastrulation-neurulation are controlled both by cell-specific and more or less stage-specific kinetic tendencies of the axially polarized cells as well as by inside-outside gradients of a rather obscure nature between the embryo or the complex cell aggregate and the ambient aqueous medium.

2. While, experimentally, individual cells from the various germ layers may exhibit tissue-specific tendencies of invagination, spreading or stretching, their movements are normally coördinated by means of a special structure, the surface coat, which unites the peripheral cells into a layer of interdependent units.

3. Although in a culture of dispersed cells from early amphibian stages any of the cells may unite indiscriminately

with any other cell, the cells later separate and sort out according to tissue specificity. Holtfreter studied these phenomena in both composite explants ('39, '44) and aggregates consisting of isolated and intermingled cells of different prospective fate ('43c). He concluded that the variety of tissue segregations, delaminations, dispersals and recombinations occurring in normal embryos could, at least largely, be attributed to stage-conditioned positive or negative "affinities" of the cells involved.

Weiss ('41, '47, '50) and Tyler ('46) suggested that selective cell adhesion involves the reaction of specific molecules at the cellular interfaces in a manner analogous to an antigen antibody complex.

The following studies are intended to further elucidate the significance of the coat, of directed cell movements and selective cell adhesion in the processes of invagination, segregation, and differentiation during early morphogenesis.

MATERIALS AND METHODS

The experimental work was performed on several species of Amphibia: *Amblystoma punctatum, Amblystoma tigrinum, Triturus torosus, Pleurodeles waltlii* and *Rana pipiens*. The eggs of the first three species were collected from sites of natural breeding in California, Tennessee, Missouri, New Hampshire and New York.[3] Fertilized eggs of *Rana pipiens* were obtained through induced ovulation and inseminated after the method of Rugh ('34). *Pleurodeles waltlii* was induced to deposit fertilized eggs in aquaria through low temperature conditioning (10°C. for 10 days) followed by injection of sheep anterior pituitary extract (Gonadophysin, G. D. Searle Co.). Most of the experiments were performed on explanted material of *Amblystoma punctatum*. The other species mentioned, and heteroplastic combinations between them, gave essentially the same results.

[3] The authors are grateful to Dr. W. W. Ballard, Dartmouth College, and Dr. V. C. Twitty, Stanford University, for egg shipments to this laboratory.

In view of the fact that the explants are highly susceptible to bacterial infection, it was necessary to employ aseptic procedures whenever feasible. The egg clusters were briefly treated with a strong solution of KOH (0.5 to 1%). This treatment has the advantage that its sterilizing effectiveness can be estimated immediately. The jelly protects the eggs for a while from being injured. If injury does occur, it can be readily noticed because of the ensuing decomposition of the eggs. Before this happens, the clusters were transferred to sterile tap water of an acid range and then washed several times. To destroy the microbes attached to the surface of the cluster, without damaging the eggs themselves, it has been found useful to observe the effect of the alkali on some eggs deprived of their protective jelly. When the latter begin to decompose, it is essential to transfer the egg clusters to a more acid medium.

The eggs were then removed from their jelly and kept for 1–3 hours in sterile tap water containing 0.5% sodium sulfadiazine and traces of streptomycin (about 20 gamma/liter; Wilde, '48). Thereafter, the fertilization membrane was removed and, after more washings, the eggs were operated on in isotonic salt solution. The loss of explants through infection was reduced to almost zero.

Methods of operation. Most of the experiments were performed on neurula material. The desired embryonic areas were excised with glass needles. In some of the experiments these fragments were simply recombined in various combinations; in others, they were briefly subjected to KOH, added dropwise to the culture medium, until pH 9.6 to 9.8 was reached. This caused disaggregation of the tissues into single cells (Holtfreter, '43a, c). The isolated cells from two or more types of tissue were then mixed with the aid of glass needles, and the alkaline medium gradually replaced by fresh solution of pH 8.0. At this lower pH the cells reaggregated.

The procedure of raising and lowering the pH requires delicate handling. If the pH rises above 10.0, excessive amounts of slime are released and some of the cells burst; re-aggregation is then difficult or impossible. However, if the limits of pH tolerance are not exceeded, the isolated cells immediately regain adhesiveness and show no signs of injury. All following experiments deal with such non-injured cell material.

Contrary to the aims of orthodox tissue culture, care was taken to avoid "outgrowth" of the explants over external surfaces. This could easily be prevented by covering the bottom of the culture dish with a thin layer of 1% agar. The culture media were Holtfreter's Solution (Holtfreter, '31) or Niu and Twitty's modification of White's Solution as reported by Flickinger ('49). The latter solution was found to be advantageous in disaggregation experiments because it is highly buffered, thereby making readjustment of pH after alkali treatment less difficult. Control embryos developed normally in this solution.

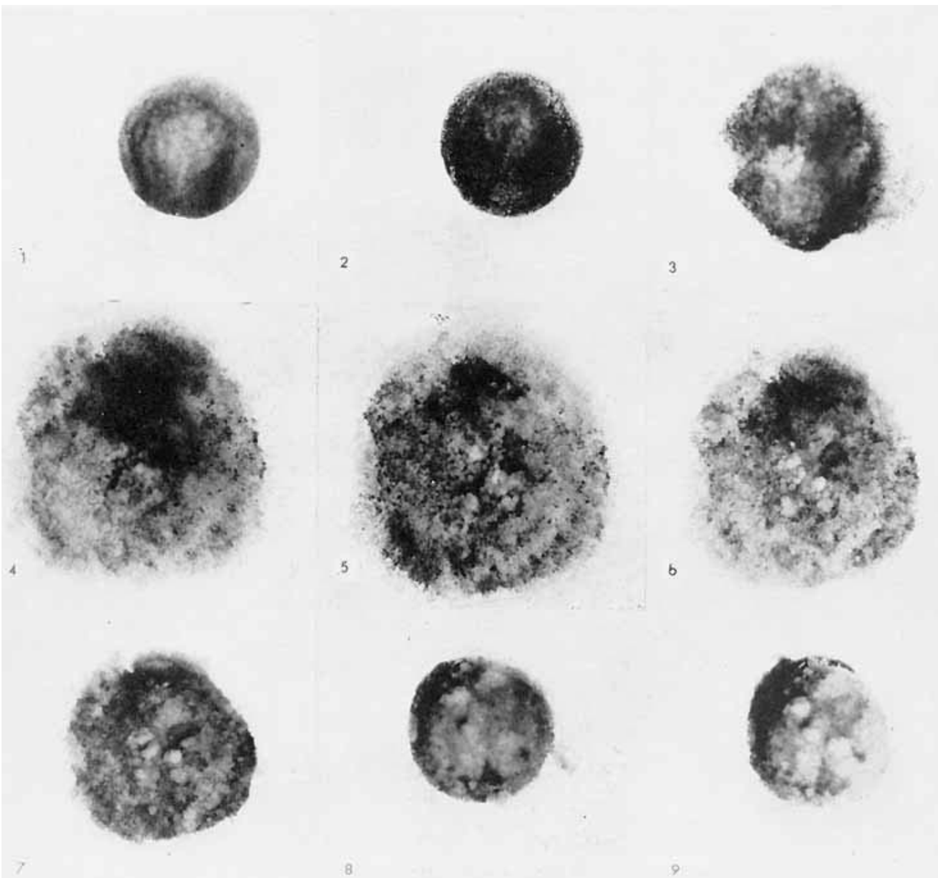
The cultured explants were examined microscopically at close intervals; concurrently representative cases were fixed for cytological study. Several thousand experimental recombinations were performed, of which 550 were studied in serial sections.

EXPERIMENTAL RESULTS

General aspects of the processes of cellular disaggregation and re-aggregation

The gross aspects of cellular disaggregation and re-aggregation are illustrated in figures 1–9, which depict this process in successive stages in a neurula which was briefly treated with alkali. Within a few minutes after a pH of 9.8 has been reached, the coated surface breaks apart. This results in the protrusion of single peripheral cells which round up and isolate themselves. Complete disaggregation occurs within some 5 minutes, at which time the neurula has been reduced to a mound of disconnected cells which are partly scattered over the bottom of the culture dish (fig. 5). When the pH of the medium is restored to the normal level (pH 8.0), the cells regain adhesiveness and form an increasingly compact mass (figs. 6–9).

Microscopic examination reveals that re-aggregation is entirely indiscriminate, i.e. any type of cell will unite with any other cell type. At this stage, therefore, cellular adhesion is non-selective. Locomotion of the free amoeboid cells plays a minor role in this process because the cells move too slowly



Figs. 1–9 A neurula of Amblystoma tigrinum, when subjected to alkali treatment, disintegrates into free cells (figs. 1–5). Following re-adjustment of pH the cells re-aggregate into a compact body (figs. 6–9).

and irregularly. Re-aggregation is mainly due to those cells which are already in point-contact with each other, and which progressively increase their areas of mutual contact until all intercellular spaces disappear.

Figures 7–9 record at hourly intervals the manner in which the cell heap regains compactness through the re-incorporation of the dispersed cells into the main body. Those cells which have not been dislodged markedly, simply re-attach themselves to the underlying cells of the broken-up neurula. The more scattered cells of the periphery undergo more spectacular morphogenetic movements. The configuration of a suspension of densely packed free elements changes first into that of a flat plate of cells; then, through further cellular condensations and re-arrangements, the peripheral cell plates are drawn into the main aggregate which eventually forms a smooth spherical body.

The most peripherally scattered cells which have lost contact with those of the central mass may fail to become re-incorporated. They tend to form spherical aggregates of their own, the number and size of which depends upon the density of the cell suspension. The loss of these cells can be prevented by simply sweeping them into contact with the main cell mass. It will be noticed — and we could substantiate this by time-lapse microphotographs — that these phenomena are much the same as those observed in the process of re-aggregation of mechanically dispersed cells from adult sponges (Wilson, '08; Galtsoff, '25, et al.).

Numerous observations made in the course of this study indicate that a temporary disaggregation of a neurula by means of alkali does not seem to change the developmental fate of the re-aggregated cells, especially not the fate of the prospective epidermal and neural cells. In other words, induction effects of an alkali treatment have not been observed in neurula material as contrasting to the observations made on temporarily disaggregated gastrula material (Holtfreter, '45; Yamada, '50). We may therefore disregard the problem of induction in most of the following considerations.

A temporary disaggregation of a neurula does not necessarily lead to grave anatomical malformations such as have been observed in alkali-treated gastrulae (Holtfreter, '48). On the contrary, when neurulae, within their fertilization membranes, were subjected to a thorough disaggregation of the ectodermal cells and then returned to normal pH conditions, the disaggregated mass of cells reconstituted itself into an intact neurula and then developed into a practically normal embryo (unpublished).

Figure 9 does not show such a return to a normal tissue configuration, simply because the dispersal of the cells from the different germ layers was carried to the extreme, namely to a total disaggregation of all cells of the decapsulated neurula. The purpose of figures 1–9 is to demonstrate that these completely dissociated cells can re-aggregate into a single mass without being pushed together by the investigator.

Combination of different embryonic cell types in explants

All of the following experiments were performed with the aim of studying the kinetic and morphogenetic phenomena subsequent to the combination of two or more well defined cell types. Two main procedures were adopted: (1) different germ layers or primordia were excised and then simply united in different combinations; (2) such different fragments were first disaggregated by means of alkali, the resulting free cells were thoroughly intermingled and then allowed to re-aggregate into a solid body (re-aggregates). We shall first discuss the experiments on the material from early neurulae (Harrison stages 13–14). The presentation will proceed from simple to more complex combinations.

A. Ectodermal derivatives

1. Combination of fragments of medullary plate and prospective epidermis. The anterior part of the medullary plate, not including the neural fold, and a piece of the prospective belly epidermis were removed and placed together with their uncoated surfaces in contact. The approximate areas excised are shown in figure 10. It may be added that in all subsequent experiments the part of the neural plate used corresponds to the area outlined in this figure.

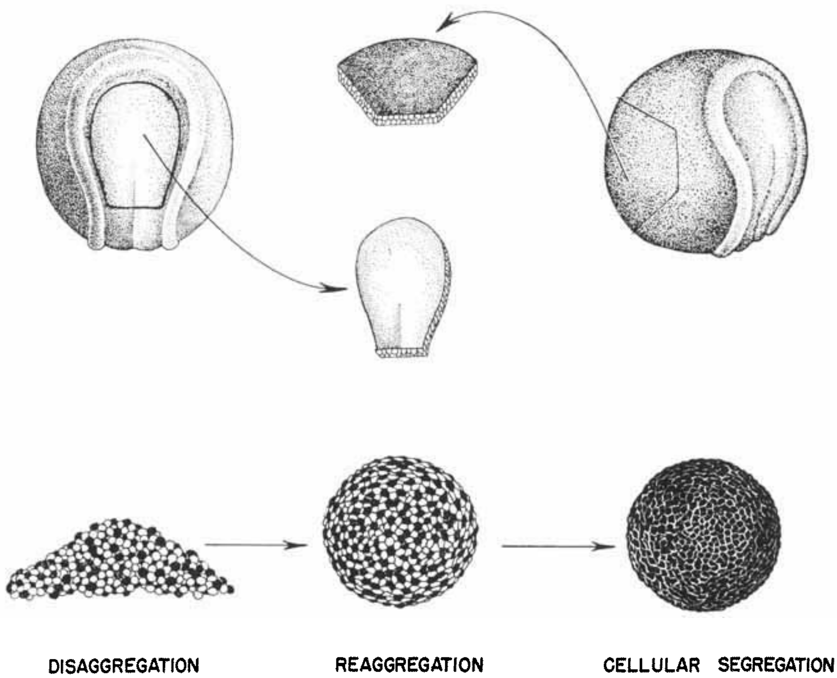


Fig. 10 A piece of the medullary plate and a piece of prospective epidermis are excised and disaggregated by means of alkali. The free cells are intermingled (epidermal cells indicated in black). Under re-adjusted conditions the cells re-aggregate and subsequently segregate so that the surface of the explant becomes entirely epidermal.

Immediate adhesion between the two tissues resulted and, within the first few hours, the medullary plate became partly enveloped by epidermis. At the same time the plate underwent an infolding similar to that which leads to the formation of the neural tube in normal development. About 10 hours later, the neural tissue, now entirely covered by epidermis, commenced to separate from the latter; this separation resulted in the formation of a space between the two tissues. After 15–20 hours the neural tissue was no longer in contact with the epidermis. This self-segregation of the tissues occurred at the same time as the detachment of the neural tube in control embryos; it is clearly the result of a newly arising lack of affinity between the tissues. Neither archenteron roof nor mesenchyme are required for the infolding nor for the separation of the neural tube from the epidermis (see also, Holtfreter, '39, '45; Barth, '41).

Sections (46 cases) revealed that differentiation of the explants was fairly typical since specific brain regions became recognizable. In all cases only a single continuous neurocoel was present. It was not our aim, as was that of Ter Horst ('48), Mangold and von Woellwarth ('50) and others, to study the capabilities of organotypical self-differentiation of the isolated different sections of the neural plate. Instead, these experiments were intended to assist in interpreting the results obtained with disaggregated cells in which the continuity of the epithelial layers was disrupted.

2. Recombination of dissociated epidermal and medullary cells. As in the preceding experiment, neural plate and ventral epidermis from an early neurula were removed. Following the procedure outlined at page 56 and illustrated in figure 10, the tissues were disaggregated and the resulting single cells re-aggregated so as to form a completely mixed composite body.

Since it is difficult to distinguish between the two intermingled cell types it was necessary to resort to heteroplastic combinations. The ectodermal cells of *Amblystoma punctatum* differ markedly from those of *Triturus torosus* in their degree of pigmentation; therefore isolated cells from the medullary plate of one species were aggregated with epidermal cells from the other species in reciprocal combinations. Xenoplastic combinations with *Rana pipiens* were also attempted, but their culture was limited to 48 hours or less, due to incompatability between the tissues. Within this period, it was seen, however, that the combined cells at first united indiscriminately and then exhibited the same shiftings that were found in homo- and heteroplastic combinations of equal age. Only the latter will be considered in the following paragraphs.

The heteroplastically intermingled epidermal and medullary cells united without delay so that within an hour all of the cells were incorporated into a single mass which was at first flat, and then, within 10–12 hours, rounded-up as indicated in figure 10. The surface of the aggregate showed a mosaic of epidermal and neural cells in a random arrangement, giving it an overall coloration intermediate to the brown and yellow tinges characteristic of the two donors. Shortly after aggregation was complete, the different cell types began the process of sorting out. At 20 hours the surface of the living aggregate acquired the smoothness of a typical homogeneous epidermis and showed the coloration of the epidermal donor. Thereafter the explants underwent a progressive swelling which commenced on the 4th day after operation and continued throughout the culture period. At the same time the surface epithelium acquired the transparency of normal epidermis and frequently was underlain by a few pigment cells probably derived from the neural plate.

Some 90 cases have been studied cytologically. They were fixed at close intervals from three hours to 12 days, with particular emphasis on the first 48 hours, since it is within this period that the major events of cellular sorting out and tissue formation occur.

The results of these and subsequent experiments are illustrated in the semi-diagrams of figures 12–26. These drawings neglect individual variations in size, shape and tissue pattern of the explants and show fewer cells than were present in the aggregates; but the essential trends of cellular re-arrangement observable in the sections are rather faithfully represented. The photomicrographs of figure 11 a–d correspond to the schematic drawings of figure 12 a–d.

During the first 10 hours the sections show closely packed, isodiametric and randomly arranged epidermal and medullary cells (fig. 12 a). Within the following few hours a remarkable regrouping begins: all of the scattered prospective epidermal cells accumulate in increasing numbers at the periphery while the medullary cells move toward the center of the explant

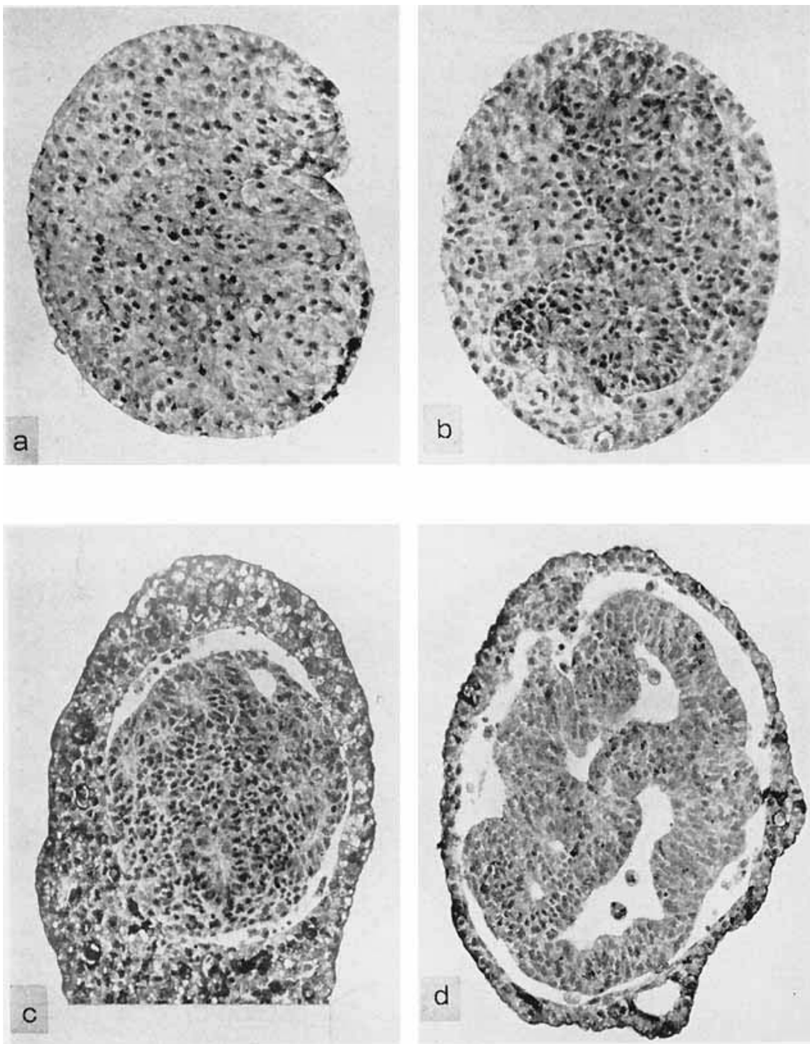
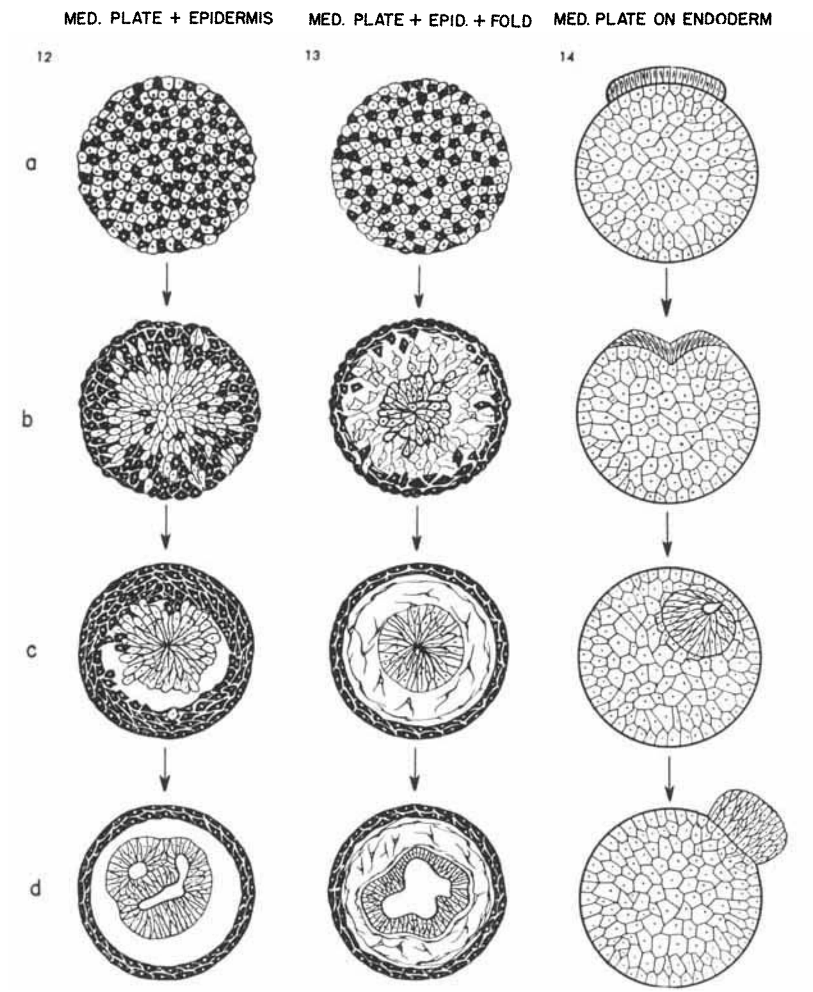


Fig. 11 Successive stages of sorting out occurring in aggregates of intermingled prospective epidermal and neural cells. The photographs correspond to the diagrams of figure 12 a–d.

where they unite into a progressively larger and more homogeneous cell mass (fig. 12 b). The boundary between the two types of cells is indistinct and jagged. Small groups of medullary cells still occupy a surface position and a number of



Figs. 12–14 Diagrammatic sections through successive stages of composite re-aggregates. Figure 12. Randomly arranged cells of epidermis (black) and medullary plate (white) move in opposite directions and re-establish homogeneous tissues. Figure 13. Same as figure 12 except that cells from the neural fold were added. The latter move to occupy the space between neural tissue and epidermal covering. Figure 14. A piece of medullary plate or of larval forebrain first moves into, then out of a matrix of endoderm.

epidermal cells may be found scattered in central locations. The centripetal migration tendency (immigration or invagination) of the prospective neural cells is reflected in their proximo-distal elongation and in the formation of pit-like depressions at the periphery (fig. 11 b). In contrast, the outermost layer of epidermal cells begins to undergo a cell-specific tangential flattening. It seems that the inward movement of the medullary cells is slightly more rapid than the outward movement of the epidermal cells since relatively more epidermal cells are found amidst the neural core than medullary cells at the periphery.

Within 14 to 20 hours after operation the sorting out becomes complete, resulting in the formation of a central homogeneous mass of medullary cells. The process of segregation continues with the appearance of a distinct and smooth line of demarcation between the neural core and the epidermal covering. Soon afterward (22–24 hours), the line changes into a cleft which demonstrates that the two types of cells are no longer adhesive (figs. 11 c, 12 c). The cleft widens and gradually encircles the entire medullary cell mass so that there is no longer any contact between the two cell types. With the expansion of the cleft into a cavity, lumina — rarely more than 5 or 6 — appear within the medullary tissue (figs. 11 d, 12 d). During the next days the epidermis undergoes a progressive expansion and thinning until, after about 4 days, a typical uniformly thin epithelium is established.

Problems of organogenesis and regional differentiation of the neural system may be briefly considered at this point. Little organogenesis results from the recombination of disaggregated medullary and epidermal cells as compared with whole fragments, even though the medullary cells form typical nuclear and fibrous layers around the lumina with differentiation into neurones. Unlike the results obtained from isolation, transplantation and defect experiments on the anterior part of the neural plate (Adelmann, '36; Ter Horst, '48; Mangold and von Woellwarth, '50) the present, more thoroughly disarranged material failed to differentiate into typical brain regions. Since the area excised as indicated in figure 10 did not contain the eye primordia, eyes did not usually develop.

As pointed out above, the temporary disintegration of a whole neurula need not lead to abnormal organ and tissue patterns of the reunited cells. The present conditions differ from those of the previous experiments in that the tissues were removed from the rest of the neurula and were thoroughly intermingled. The absence of typical regional organogenesis in the present re-aggregates may be ascribed (1) to the absence of a continued determinative influence of the archenteron roof, and (2) to the fact that the neural plate, at the time of operation, may be considered as determined in a local, though rather vaguely circumscribed sense, and that a haphazard shuffling of its cells cannot be corrected by way of regulation to restore typical organ pattern. At any rate, these results show that this medullary material (fig. 10) does not represent a single embryonic field which is capable of reconstituting itself after a thorough disarrangement of its constituent cells.

Whatever relations may exist between the field properties of the induced parts of the neural plate and their subsequent shaping by the underlying mesoderm, the present data bring out clearly two other principles, namely cell-specific movements and adhesions. Their combined effects result in tissue segregation. Two conclusions are indicated by these results: (1) The medullary cells need not constitute a continuous coated epithelium in order to invaginate. Each individual cell has this tendency. (2) A central lumen which normally results from the infolding of the plate, can be formed equally well by way of secondary cavitation.

3. Recombination of dissociated cells from medullary plate, neural fold and prospective epidermis. The material consisted of a single sheet of ectoderm comprising the anterior portion of the plate plus adjacent neural fold and lateral epidermis which was isolated from early neurulae of *A. punctatum.* Disaggregation and re-aggregation as before (20 sectioned cases).

The sorting out of medullary and epidermal cells progressed as in the preceding experiment, with inward movement of the former and outward movement of the latter. The cells derived from the neural fold became located between the accumulating mass of centrally located medullary cells and the re-established layer of peripheral epidermis (figs. 13 a–d). The end-result was a perfectly neat segregation of the three types of tissues which attained a concentric arrangement similar to that in normal embryos.

The addition of neural fold material did not lead to neural organ patterns superior to those of the preceding experiment. According to prospective significance, the cephalic neural fold gave rise to mesenchyme, melanophores, small ganglionic cell clusters and a corium underlying the epidermal covering. Contrary to prospective significance, no cartilage was differentiated. This may be attributed to the absence of pharyngeal endoderm which, according to several authors, is necessary to transform cephalic mesectoderm into cartilage (Hörstadius and Sellman, '46).

As in the preceding experiment, the newly established core of medullary cells formed a variable number of neurocoels by way of cavitation, and the tissue underwent differentiation into mantle and marginal layers. The neurocoels tended to become excessively expanded and their configuration provided no criteria to compare them with normal divisions of the brain. However, some abortive eye-like formations consisting of accumulations of retinal and tapetal cells were found.

The essentially new problem in this experiment is to find an explanation for the rearrangement of the cellular derivatives of the neural fold in such a way that they finally settle between epidermis and neural tissue.

B. *Ectodermal and endodermal derivatives*

1. Combination of medullary plate and endoderm from a neurula. The isolated anterior part of the neural plate and a mass of ventral endoderm of early neurulae were combined. The two tissues adhered to each other without delay, but the originally flat piece of neural plate curled up rapidly, thus anticipating its normal infolding process.

The plate subsequently flattened out again, increased its contact surface with the endoderm and then sank as a contracted mass into the endoderm. After about 10 hours most of the medullary tissue had disappeared from the surface and, some 10 hours later, it had been engulfed entirely by the endodermal matrix (fig. 14 a–c).

In contrast to the movements of medullary cells into an epidermal matrix (p. 64), or of mesodermal cells into the very center of an endodermal mass (p. 83), here inward movement was less complete. The invaginated dark cell mass could be seen shining through a thin layer of endoderm cells which formed a daisy-like pattern, observed in other instances of invagination and wound healing (Holtfreter, '43a, '44).

After 30 hours the rounded medullary fragment began to reverse its direction of movement and return to the surface (fig. 14 d). The whole mass re-appeared within 40–45 hours. Obviously, the initial incorporation and subsequent expulsion reflect changes in affinity between the two types of tissues.

Examination of the sectioned material (28 cases) revealed more details concerning cellular arrangement and lumen formation. When the piece of medullary plate had slipped half-way into the endodermal matrix, it exhibited the U-shape infolding, and elongation of its cells, characteristic of normal neurulation (Lehmann, '26; Boerema, '29; Gillette, '44). But when invagination was complete, the neurocoel disappeared entirely, soon after it had been formed by way of infolding. Explants fixed at successive intervals showed a progressive fusion of the walls of the neurocoel until only the accumulation of pigment indicated a former cavity. Eventually the invaginated plate material formed a solid mass of cells that returned without change to the periphery of the endodermal matrix.

Apparently, the surface coat which is responsible for the maintenance of invaginated cavities in general (Holtfreter, '44, '48), becomes dissipated under the present conditions, just as the coat of the mesodermal primordia disappears during the course of normal invagination. This disappearance of a neurocoel differs from the results of the preceding experiments where the experimental conditions led to the re-establishment of neurocoels within a compact mass of prospective neural cells. It is obviously the abnormal endodermal environment which inhibits the formation of neurocoels.

With the loss of an inner cavity, the medullary cells more or less lose their radial arrangement but they continue differentiating into neural elements. The finally expelled cell mass remains slightly attached to the endoderm with its fibrous layer, whereas the nuclear layer, showing a ragged surface, is turned toward the external medium. Within 9 days of culture there occurred no infiltration of nerve fibres into the underlying aggregation of large and undifferentiated endodermal cells.

The time required for complete invagination into the alien matrix of endoderm is about the same (some 20 hours) as that for transformation of the neural plate into a tube in control embryos. Thus the kind of histological environment does not seem to influence specifically the rate of invagination. However, the endodermal environment proves to be unsuited in that it prevents neurocoel formation and does not permit a prolonged sojourn of neural material within this cell mass or the outgrowth of nerves into it.

The following experiments show that endoderm from a wide range of stages can equally well serve as a matrix for the invagination of medullary material.

Lack of stage-specificity of the endodermal substratum. When fragments of medullary plate were combined with isolated endoderm from blastula and gastrula stages (17 sectioned cases), it was found that the time sequence of events — invagination followed by expulsion — was the same as described above with reference to endoderm from the neurula.

In another experimental series (9 cases), medullary plate was combined with endoderm from advanced stages, namely from stages 13, 14, 19, 22 and 33. In this series, the endoderm was not isolated but simply denuded by removing the overlying epidermis and mesoderm of the lateral trunk region. A piece of medullary plate was then placed upon the smooth endodermal surface. Hosts older than stage 33 proved to be unsuited for this experiment because the graft would not adhere to the endodermal surface. This lack of adhesiveness possibly illustrates the change of affinity which has been mentioned above.

When the piece of medullary plate did adhere to the denuded, aged endoderm, it underwent the processes of initial invagination and subsequent expulsion as described above. Only when the advanced host stages 22 and 33 were used, complete invagination was rarely achieved. There was also some slight delay in the initial attachment of the medullary material to the endoderm of these older embryos.

It follows from these observations that the age of the endodermal substratum does not affect the movements of the medullary material. At any rate, it is not just the endoderm of a neurula which permits invagination of a piece of medullary plate.

As a further instance of the unspecificity of the substratum and its developmental stage, a fragment of medullary plate readily invaginates into the interior of a morula. In these instances (15 cases) the graft slipped in between the large cells of the animal pole region to become permanently lodged within the underlying endoderm.

Lack of stage specificity of the invagination tendency. The preceding experiments demonstrate that it is not the specificity of a certain germ layer nor the age of the cellular substratum which control the movement of medullary tissue into the depth. The following experiments were performed in order to examine whether or not the tendency for invagination is specific only for the medullary plate stage.

Excised fragments of larval forebrain (stages 23–30) were brought into contact with isolated endoderm from a neurula (9 sectioned cases). The fragments adhered immediately, though not very firmly to the much younger endoderm. Within 15 minutes, adhesion was sufficient to resist mechanical separation of the tissues. The brain fragment moved into the endodermal substratum and, within 20 to 33 hours, entirely disappeared from view. During the second day it returned to the surface where it remained for two days, barely attached to the endoderm.

Thus, with respect to directed movements, the larval forebrain behaved essentially like medullary plate. These processes were executed at practically the same time intervals in all of the experimental modifications investigated.

Whereas in the preceding instances the neural material moved inward either by way of infolding (invagination) of a plate of cells or by way of immigration of scattered cells, the piece of larval brain slips into the endoderm as a solid lump of cells ("ingression") and there is no interdigitation between the neural and endodermal cells. One might wish to find a common mechanism for these three modifications of inward movement.

Additional experiments (9 cases), in which larval brain was combined with endoderm beyond the neurula stage (stages 19–22), merely confirmed this conclusion: within the range of the stages investigated, neither the age of the embedding endoderm nor that of the neural graft influence either the mode or the time sequence of invagination and subsequent expulsion. If invagination were simply a function of the state of differentiation of the two tissues, one would expect that the neural tissue from older donors, which has already undergone invagination and segregation, would fail to adhere to younger endoderm and to invaginate and segregate itself once more. Instead, the results show that neural tissue and endoderm from a wide range of stages have the tendency to establish at first a maximal mutual contact for a definite period, and then to separate, irrespective of the age of either component. Normal neurulation seems to result from a fortunate coincidence of invagination tendency and a favorable matrix.

These findings warrant a revision of the theoretical schemes devised to account for the change of affinity between different tissues. Apparently in cellular adhesiveness, two principles are involved: a non-specific adhesiveness that is found in early embryonic stages and persists in older material, and a cell-specific adhesiveness that develops after the different cell types have been in contact for a definite period. This would imply that selective cell adhesion is only partly the result of aging of the cells concerned. It is also the result of mutual interactions between the combined cells of different germ layers.

In the following experiments we return to the earlier problem of the significance of the coat in invagination and of the effects which different tissue environments exert upon the shaping of tissue patterns.

3. Combination of endoderm with neural plate and neural fold. When the neural fold was added to combinations of isolated, non-disaggregated medullary plate and endoderm (6 cases), the medullary plate invaginated into the endoderm and most of the neural fold material remained close to the surface differentiating into epidermis at the periphery and into mesenchyme underneath and between neural tissue and endoderm (fig. 15 a–d). The mesenchyme acted as true binding material connecting neural tissue with epidermis as well as with endoderm. No expulsion of neural tissue occurred. The presence of epidermis-covered mesenchyme permitted the neural tissue to maintain a central lumen. Here again, regional specification of the neural formations has not been observed.

4. Combination of dissociated medullary plate and endodermal cells. Anterior medullary plate and ventral endoderm of the early neurula (stage 13) were isolated, disaggregated and re-aggregated. The difference in size between the two cell types promotes some clustering of the medullary cells; however, once aggregation has started, this preliminary grouping ceases.

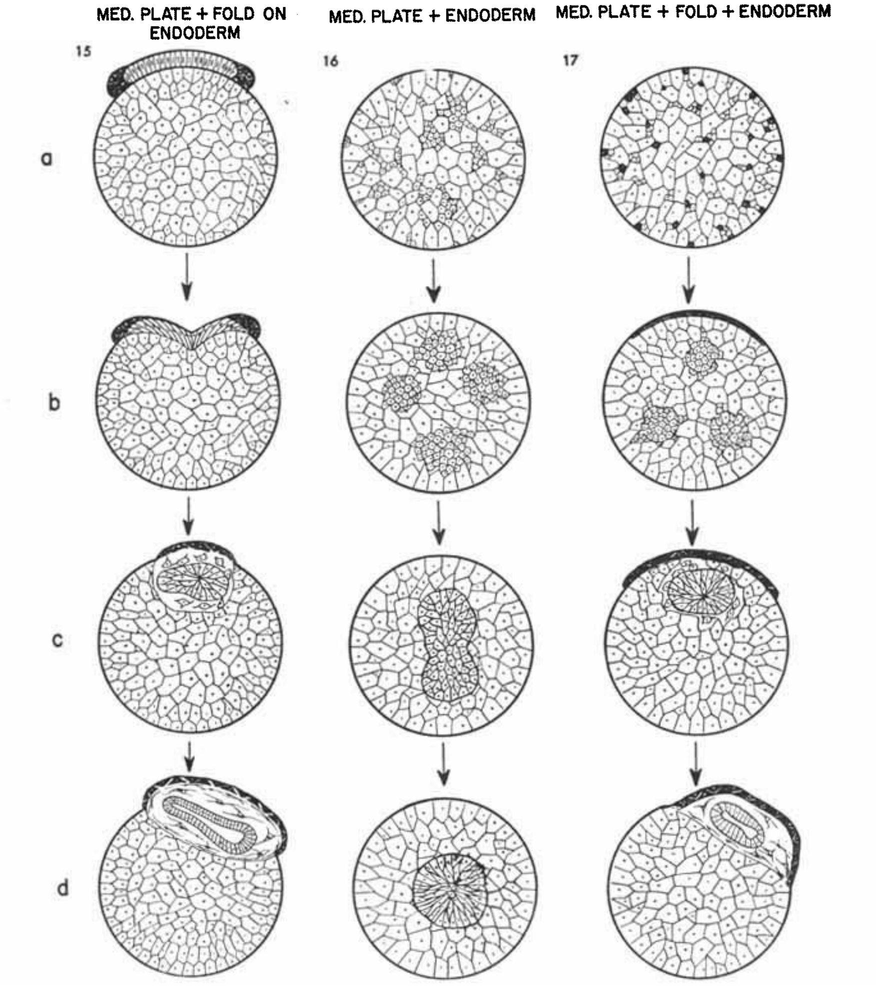


Fig. 15. When neural fold and medullary plate are combined with endoderm, invagination of the medullary material is incomplete and the developing epidermis and mesectoderm prevent isolation of the neural tissue. Figure 16. When dissociated cells of medullary plate and endoderm are mixed, the former move centripetally to produce a core of neural tissue lacking a neurocoel. Figure 17. The addition of neural fold produces epidermis and mesenchyme which prevent central allocation of the neural tissue and promote neurocoel formation.

Within three hours after operation the aggregate becomes a smooth, more or less spherical mosaic of endodermal and neural cells. The latter invaginate rapidly so that, after 14 hours, few if any remain at the surface. The surface aspects change very little after the first 20 hours, except for an occasional thinning of the endoderm overlying the neural tissue. In some, though infrequent instances, the endoderm may recede, exposing the neural mass to the medium.

The dynamics of the segregation process are more clearly revealed in sections (37 cases). Five hours after re-aggregation one finds single medullary cells or small clusters of them interspersed in the gross texture of the endoderm cells (fig. 16 a). The peripheral medullary cells are still invaginating, as may be deduced from their formation of pits and from their elongation perpendicular to the surface.

Intrinsic cell polarity controls the direction of movement. According to observations of Holtfreter ('47) on isolated cells, it is the inner, non-coated portion of the elongated medullary cell which forms the anterior pole in amoeboid locomotion. In the individually scattered medullary cells within the present aggregates, it is this cell pole which can be seen to turn toward the center of the aggregate while the originally coated and still heavily pigmented distal pole becomes turned distally. Thus the centripetal cell movement seems to be due to a response of the kinetically most active proximal cell portion to an inside-outside gradient of some kind existing within the aggregate. It is, however, difficult to apply the notions gained from the study of isolated cells directly to the solid clusters of medullary cells which likewise move into the depth within the first 5 to 24 hours (fig. 16 b, c).

After 24 hours not more than two medullary clusters were found and, by 40 hours, all of the originally dispersed cells became united into a single central mass which was clearly demarcated from the surrounding endoderm by differences in cellular shape, size and orientation with a distinct line of pigment at the outside of the medullary cell mass. As in the case of the combination of dissociated medullary and epidermal cells, a neat sorting out of the two cell types takes place.

The centrally assembled prospective neural cells formed a homogeneous tissue which failed to undergo regionally specific differentiation. Within this mass, the cells became arranged in irregular whorl-like patterns — an abortive attempt at organizing themselves. As in the preceding experiments, the embedding endoderm proved to lack the proper conditions for the formation of neurocoels which occurred when reconstitution of neural tissue took place within an epidermal covering (p. 66).

In contrast to the experiments with whole pieces of medullary plate (p. 69), the neural tissue did not return to the surface of the embedding endoderm during the culture period of 10 days. To attribute this to the absence of a continuous coat is not convincing in view of the fact that a piece of forebrain, lacking a coated surface, does move back to the endodermal surface. Further studies must elucidate this problem.

5. *Combination of dissociated cells from medullary plate, neural fold and endoderm.* When neural fold material is added to the cell mixtures considered in the preceding paragraph, the neural cells invaginate as described above (p. 75). At 20 hours the surface of the aggregate consists of patches of endodermal and ectodermal cells. The latter spread and fuse with each other until, on the 5th day, there is usually a single cap of epidermis which becomes elevated and finally underlain by pigment cells.

In sections (6 cases), segregation of neural and endodermal cells proceeds as described previously, when neural fold was not included. During this process the neural tissue becomes surrounded by the differentiating mesenchyme, which is evidently responsible for the fact that well formed, though irregularly shaped lumina arise in the neural tissue. The final distribution is principally the same as that which results from a combination of corresponding intact fragments (compare figs. 17 and 13).

These data give further support to the notions gained from the preceding experiments. They show clearly that in the process of sorting out, the different cell types exhibit a cell-specific tendency to arrange themselves in a definite tissue pattern which corresponds to that in normal development.

6. Combination of fragments of prospective epidermis and endoderm. Fragments of prospective epidermis and ventral endoderm from the gastrula and neurula (stages 10 and 13) were combined. The results obtained were consistent irrespective of the different age of the tissues employed (13 sectioned cases). Detailed description of this experiment is not necessary because similar tissue combinations have been described by Holtfreter ('39, '44) whose results we could confirm.

A piece of prospective gastrula or neurula epidermis whose inner surface has been brought into contact with non-coated endoderm, adheres to the latter and then spreads peripherally. This movement is carried out, not because of the contractile strength of the coat (Lewis, '47), but in spite of it; it results from an active amoeboid spreading of the whole sheet of epidermal cells. The basal part of the endodermal ball which touches the agar-plated culture dish does not become covered by the spreading ectoderm. Similarly, in whole embryos which are kept in standard solution on a glass surface, the epidermis which touches the glass, tends to recede from the embryo and to spread sheet-like over the glass; but such extra-embryonic spreading will not take place when the medium is simply water or when the embryo rests upon agar.

For about 24 hours the epidermal cap continues spreading and flattening centrifugally over the body of endoderm. Then epiboly ceases and, after 4 days, the epidermis reverts to the opposite movement of contracting and receding from the endoderm. This signals the onset of a state of disaffinity between the two tissues. For the following two days the epidermal cap becomes progressively thicker, more wrinkled and less attached to the endoderm. However, complete self-isolation of the two tissues components, such as has been reported by Holtfreter ('39), was not observed within the culture period of 10 days, although there remained only sporadic points of mutual adhesion. Holtfreter cultured his preparations for a longer period.

7. Combination of dissociated prospective epidermal with endodermal cells. When the same tissues as used in the preceding experiment were briefly subjected to alkali and then allowed to re-aggregate haphazardly, they formed a firm, compact mass within 2–3 hours (fig. 18 a). The surface of the mass was at first a mosaic of epidermal and endodermal cells except that the unequal size of the two cell types promoted some clustering of the epidermal cells. The peripheral epidermal cells immediately began to spread over the endodermal cells indicating that the former have a stronger spreading tendency than the latter. The internally buried epidermal cells remained isodiametrical.

As seen in sections (14 fixed cases), the cellular sorting-out proceeds rapidly so that, after 12 hours, all of the internally located epidermal cells become linked up with the peripheral ones by radiating cellular connections. Within the next 10 hours or so, all epidermal cells have moved to the periphery where they spread and unite to form an epidermal covering. If the number of epidermal cells is insufficient to cover the whole endoderm, or if the contact between aggregate and culture dish prevents epidermal spreading over the basal portion of endoderm, the condition illustrated in figure 18 c results, namely an epidermal "capping." From the 4th day on, the cap contracts and detaches itself progressively from the endoderm as a wrinkled mass (fig. 18 d). Complete self-isolation of the two tissues seems to be only a matter of a sufficiently long culture period.

To conclude: the initial indiscriminate union between dissociated and intermingled prospective endodermal and epidermal cells is soon followed by their sorting-out through directed cell movements and selective adhesions leading to the formation of homogeneous layers of superficial epidermis and internal endoderm. Both the ectoderm and endoderm are obviously capable of establishing peripheral epithelia, but when competing with each other, the ectodermal cells display a stronger tendency than the endodermal cells to move

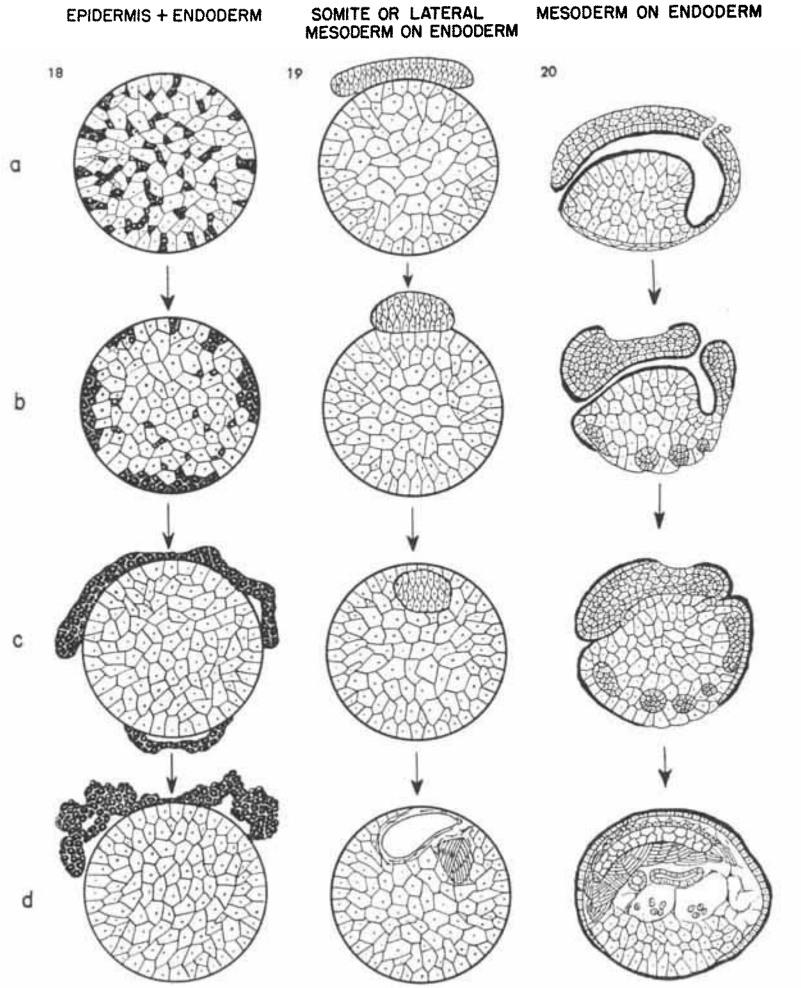


Fig. 18 The recombination of dissociated epidermal and endodermal cells leads to a sorting out and self-isolation of homologous tissues. Figure 19. Combination of a fragment of the mesoderm mantle with trunk endoderm results in the incorporation of the mesoderm into endoderm. Figure 20. Removal of the whole ectoderm of a neurula or older stages leads to a movement of mesoderm into endoderm and a spreading of the coated endoderm of the archenteron over the denuded embryo (coated surfaces indicated by a heavy black line).

peripherally and to spread as a surface epithelium. However, a permanent association of the two layers appears to be impossible for the lack of connective tissue.

C. Mesoderm and endoderm derivatives

1. Combination of fragments of prospective notochord and endoderm. Prospective notochord and ventral endoderm were excised from neurulae (stage 15) and placed in contact. Firm adhesion occurred within 5 minutes. The notochordal tissue invaginated much faster than did neural tissue, namely within 2–3 hours. A thin layer of endodermal cells shifted over the graft, thereby exhibiting the characteristic daisy pattern.

Sectioned material fixed after 24 hours (9 cases) showed the notochordal tissue as a round compact mass lying close to the surface and being sharply delimited from the endoderm. Typical vacuolization and a distinct sheath became evident on the third day. During the next few days the notochord elongated and in so doing, a terminal tip would sometimes break through the endoderm. The two tissues remained in close contact for as long as 9 days, during which time the notochord differentiated normally except for the fact that its elongation was not as extensive as in intact embryos. Thus, in contrast to the preceding combination, no disaffinity arose between notochord and endoderm during the period under examination.

2. Combination of fragments of prospective somites or lateral mesoderm with endoderm. Prospective somites or lateral mesoderm from a neurula when placed on ventral endoderm were found to behave much like prospective notochord (figs. 19 a–d). The mesoderm upon isolation contracted and proceeded to invaginate into the endoderm within a few hours. Differentiation of these tissues (11 cases) was according to prospective fate: the dorsal mesoderm formed typical somites, while the hypomere produced coelomic cavities within the endoderm. These aggregates showed no sign of self-isolation of the different tissue components for as long as 10 days.

3. Combination of the total mesoderm and endoderm of a neurula. The experiments reported above demonstrate that different parts of the mesoderm mantle will invaginate into isolated endoderm and undergo typical differentiation. We shall presently consider the kinetic behavior of the entire mesoderm mantle of a late neurula (stage 20) when left in contact with the denuded endoderm. To obtain such preparations the entire epidermis and neural plate were removed (fig. 20 a). At stage 20 the mesoderm is clearly demarcated which permits this operation without injury to the archenteron roof (12 sectioned cases).

Following removal of the ectodermal surface layer, the chorda-mesoderm mantle contracts and moves into the underlying entoderm. Its dorsal portion invaginates as a compact mass, whereas its latero-ventral portion breaks up into patches which slip independently into the endodermal matrix (fig. 20 b, c). After 10 hours only endoderm is present on the surface while the still undifferentiated, entire mesoderm collects as a single, centrally located mass of cells. This mass undergoes histotypical differentiation into notochord, somites, pronephros, coelomic cavities, blood cells, etc. which are rather haphazardly arranged (fig. 20 d). A more typical bilateral and antero-posterior alignment of the axial mesoderm occurs when older embryos (stage 28) are deprived of their ectodermal covering.

It should be pointed out that the endoderm does not merely play a passive role in the process of transportation of the mesoderm into the depth. Any kind of invagination involves movements both of the invading and the embedding cell material. Under the present conditions, the coated endoderm of the archenteron (indicated by a heavy black line) moves out through the slit-shaped blastopore, or through an accidental hole in the archenteron roof, and spreads actively over the denuded mesoderm and uncoated endoderm (fig. 20 b, c). The whole preparation thus becomes turned inside out, i.e. the coated endoderm of the archenteron eventually becomes the external covering of the entire preparation. Such inversions and epibolic expansions of coated over non-coated endoderm (and mesoderm) have been described before (Holtfreter, '43a, '44).

The general configuration attained in this experiment resembles that of a total exogastrula in which an endodermal instead of an epidermal epithelium forms the envelope of the mesodermal tissues. Here, however, the inversion of the two germ layers takes place *after* they had already achieved normal topographic arrangement. These observations permit two general conclusions: (1) An ectodermal covering seems to be the necessary prerequisite for the normal shifting of the mesoderm over the endoderm as well as for the maintenance of the mesodermal sheet; in the absence of an ectodermal covering, any part of the mesoderm will exhibit the "abnormal" behavior of slipping into the interior of an adjacent endodermal cell mass. (2) The tendency of the mesoderm to invaginate is not restricted to a certain stage, which is normally the gastrula, but it is present, though normally not used, in neurula and probably older stages.

4. Combination of dissociated cells from the chorda-mesoderm and endoderm of the neurula. The entire complex of endoderm-mesoderm tissues employed above was disaggregated, mixed-up and allowed to re-aggregate. As in all previous re-aggregations, initial union of the cells was at random and a spherical compact mass resulted in a few hours (fig. 21 a). The mesoderm cells exhibited a pronounced invagination tendency so that after the first 5 hours they were no longer visible at the surface. In addition to the large endodermal cells from the trunk region, a number of much smaller prospective foregut cells collected on the surface in small patches. After 20 hours these patches became more conspicuous for they began to differentiate into protruding bulges of epithelium. The prospective ventral endoderm could also form peripheral membranous epithelia. In later stages of culture (10 days) many of the explants exhibited cardiac pulsations which were easily detected if they occurred underneath the flexible foregut endoderm.

In the sectioned material (39 cases) it is apparent that at 9 hours all of the mesodermal cells have left the surface and are scattered throughout the endoderm in the form of single cells or aggregates of various sizes. Within the next three

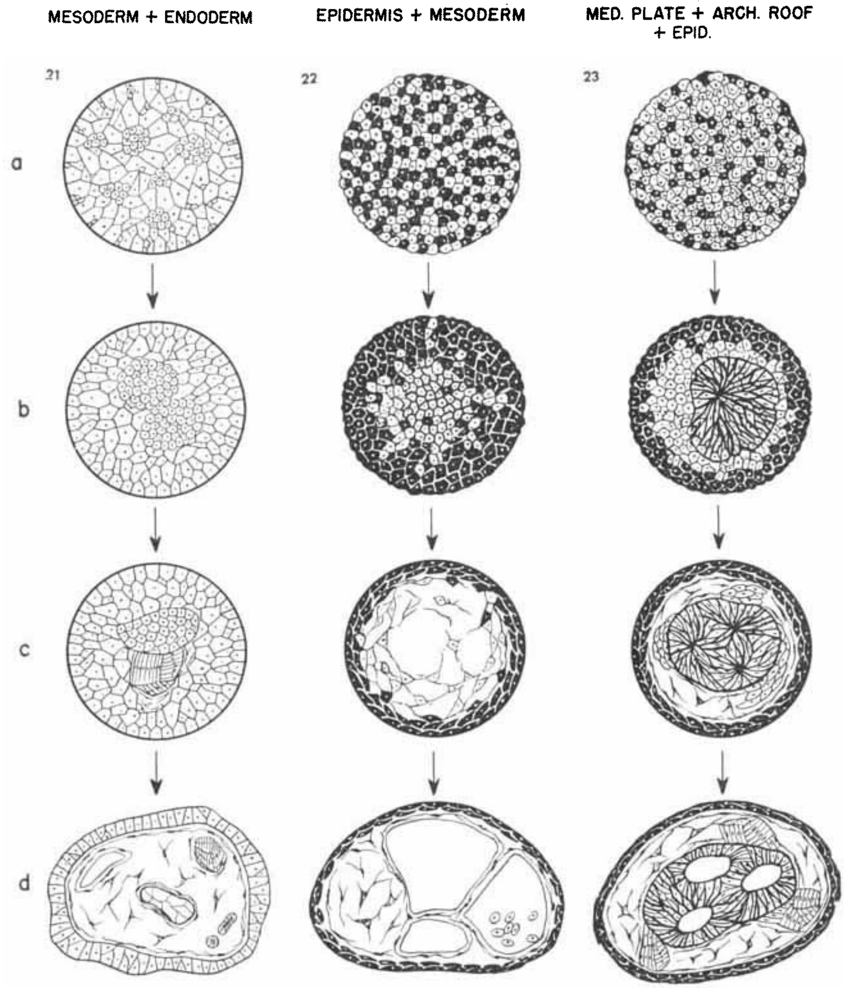


Fig. 21 Combination of dissociated mesoderm and endoderm cells results in centripetal migration of the mesoderm cells. Figure 22. Outward movement of epidermal cells and inward movement of mesodermal cells, the latter eventually forming mesenchyme, coelomic cavities and blood cells. Figure 23. Sorting out of randomly arranged cells from medullary plate, epidermis and axial mesoderm resulting in the formation of centrally located neural tissue which is surrounded by somites, mesenchyme and epidermis.

hours the number of aggregates decreases and their mass increases. At 35–44 hours only two mesodermal aggregates are found (fig. 21 b) and, after 50 hours, they have united to form a single central mass (fig. 21 c). Notochordal and somite differentiation becomes recognizable after 90 hours, while much of the remainder of the mesoderm is still relatively undifferentiated.

On the 5th day of culture the somites exhibit characteristic segmentation while notochordal tissue becomes vacuolated. A number of mesenchyme lined cavities also appear and can be designated as coelomic cavities, particularly since the pronephros is associated with them. Large numbers of loose cells — probably early blood elements — are found in the cavities. In addition, spaces not lined by mesenchyme develop within the endoderm, due to cavitation. Pronephros becomes well differentiated on the 6th day. The pronephric tissue is usually a single, continuous, twisted tube, but in some instances additional tubes are present which are not continuous with the main tube. In older preparations one finds well differentiated gut epithelia, which are peripherally located and usually underlain by mesenchyme as shown in figure 21 d. The heart fragments which were noticed because of rhythmical contractions could not be clearly identified in sections. A bilateral and axial disposition of the re-established tissues fails to develop in such aggregates.

D. Ectoderm and mesoderm derivatives

1. Combination of dissociated cells from the latero-ventral epidermis with mesoderm. Latero-ventral epidermis and underlying mesoderm were removed from neurulae (stages 13–14) and dissociated into single cells which were recombined. As a result of directed cell movements within the aggregate, an epidermal surface was established after 10 hours while the mesodermal cells slipped inside. For the most part the latter left the surface before the aggregate became compact. The explant swelled markedly and changed from a solid ball to a vesicle covered by a transparent epidermis. Melanophores sometimes appeared after 7 days; their origin is problematical. Since the neural fold was not included, it appears that they have arisen from the area immediately ventral to the neural fold, or by induction. Rhythmical contractions suggesting the presence of heart fragments were frequently observed.

In sections (18 cases) the results were as follows. Cell segregation, effected through the inward and outward movements of the mesodermal and epidermal cells respectively, eliminates the mosaic condition (fig. 22 b) within the first 10–15 hours. The explants are then composed of a solid core of mesoderm surrounded by a thick shell of epidermis. During the following 10 days the mesoderm which initially consisted of tightly packed isodiametric cells, becomes differentiated into blood elements, vessels, mesenchyme and corium, the latter underlying the thinned-out epidermis (fig. 22 c, d). Occasionally a few pronephric tubules may be seen; however in most cases the presumptive pronephros was not included in the tissues excised.

To conclude: the opposite movements of the re-aggregated mesodermal and epidermal cells result in a segregation of the two cell types and in the formation of a vesicle such as develops when the same ventro-lateral material is simply explanted as a fragment.

2. Combination of dissociated cells from the medullary plate, archenteron roof and epidermis. Excised anterior part of the medullary plate together with underlying archenteron roof and a piece of ventral epidermis were dissociated and the free cells re-aggregated after mixing (21 sectioned cases). Shortly after removal of the alkali, all cell types firmly adhere to one another (fig. 23 a). During the first 20 hours the surface of the explant becomes completely epidermal. This condition is brought about through the inward movement of the mesodermal and neural cells as well as the outward movement of the epidermal cells. This results in the three-layered configuration of a central ball of neural tissue, surrounded by a mantle of mesoderm, which in turn is enveloped by a shell of epidermis (fig. 23 c). Obviously, although both the neural and mesodermal cells have an invagination tendency, it is more pronounced in the former than in the latter.

During the following 4–5 days the epidermal surface commences to bulge out due to subjacent mesenchymal differentiation. Within 8–10 days the neural mass develops a number of neurocoels — as many as 6 may be present in a single complex — which makes it appear brain-like (fig. 23 d). Identification of distinct brain divisions is not possible, but cytological differentiation proceeds well, with fibres being produced. Abortive eye cups, or secondary induction products e.g. frontal glands, small ear vesicles or nasal placodes may appear.

The mesoderm forms scattered mesenchyme, irregular masses of somites, and notochord tissue, generally closely associated with the neural tissue. Although there is usually a continuous single notochord, isolated fragments of it are occasionally found trapped in the neural tissue. The mesodermal tissues, though differentiating normally, fail to adopt a typical bilateral and axial arrangement. This may be partly the reason for the absence of typical organisation in the adjacent neural tissue.

E. Ectoderm, mesoderm and endoderm derivatives

1. Combined fragments of medullary plate, chorda-mesoderm and endoderm. Head plate plus underlying chorda-mesoderm were combined with ventral endoderm of the early neurula. The grafts invaginate into the endoderm within the first 2–3 hours and, in so doing, become folded into a "U" configuration (fig. 24 a–c). After 12 hours, invagination is complete but during the subsequent 12 hours the movement is reversed. This results in the reconstitution of the open-book-shaped plate, a condition which is attained after 42 hours and remains for at least 6 days (fig. 24 d). Notochord and somites differentiate quite typically.

It is rather surprising that the graft is expelled in spite of the presence of mesoderm which, when grafted alone, does not experience this fate. The mesoderm remains firmly attached both to the endoderm and neural tissue. Evidently it

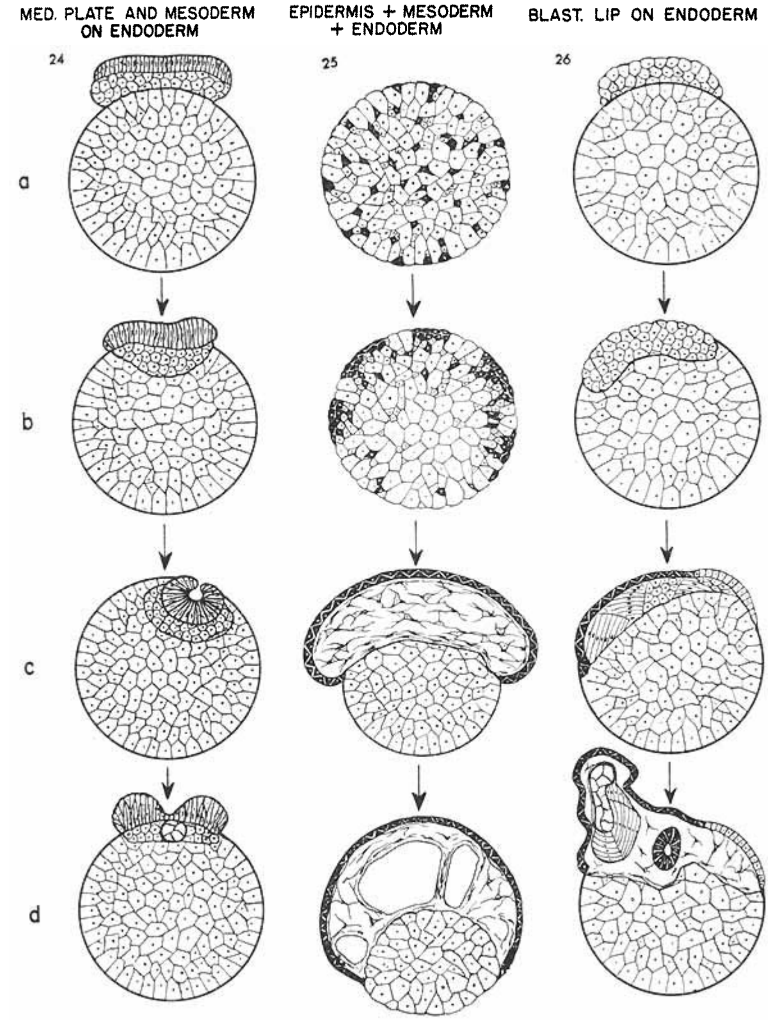


Fig. 24 Incomplete invagination of medullary plate and axial mesoderm when placed on trunk endoderm. Figure 25. Cell segregation and tissue formation which occur in re-aggregated cell mixtures of epidermal, mesodermal and endodermal cells. Figure 26. A fragment of the blastoporal lip only partially invaginates into a socket of endoderm. It forms a peripheral epithelium of epidermis (black) and fore-gut endoderm; underneath axial tissues and a neural tube.

is the emerging disaffinity between neural tissue and endoderm which drives the graft to the surface. The strong adhesion between chorda-mesoderm and neural tissue, in conjunction with the absence of epidermis, causes the neural material to form a spina-bifida instead of a tube.

2. Combination of dissociated cells from the latero-ventral mesoderm, epidermis and ventral endoderm. The ventral and lateral epidermis plus underlying mesoderm and trunk endoderm were removed as a whole (stage 13–14), disaggregated and allowed to re-aggregate. The amount of endoderm which was included varied from the whole ventral mass to small fractions of it. Depending on the relative amount of endoderm, one of two things happened: If the entire endoderm was used the epidermis and mesoderm were unable to completely cover it; with smaller amounts of endoderm it became entirely covered. Our discussion will be limited to the first condition (20 sectioned cases).

Within 10 hours after re-aggregation all of the epidermal cells accumulate peripherally. Although several patches of epidermis may be present initially, they soon fuse to form a single cap. The endoderm facing the bottom of the culture dish remains uncovered by epidermis. While some mesoderm may be buried in the endoderm, it becomes mainly located between epidermis and endoderm (fig. 25 a–d). Mesenchyme, pronephros, blood cells, and mesothelium appear in late stages of culture. A small number of trapped cells may be found: single endodermal cells in the mesenchyme, or a few mesodermal cells in the endoderm. The epidermis is nowhere in direct contact with endoderm.

The final tissue configuration is essentially like that of the normal embryo. Its formation and maintenance were possible because: (1) the presence of ectoderm prevented the mesoderm from accumulating in the center of the endoderm; (2) the presence of mesoderm prevented the ectoderm from self-isolation.

3. Combination of endoderm with prospective chorda-mesoderm from an early gastrula. Heretofore we have been concerned with tissues and cells of neurulae and older stages which at the time of recombination were for the most part determined and had already undergone the morphogenetic movements of gastrulation. It was the more surprising that under these conditions several of the tissues again executed the movements characteristic of the gastrula stage. The following experiments deal with material from the early gastrula, the developmental fate of which is not at all locally determined. The material used consisted of most of the prospective areas of chorda-mesoderm and cephalic endoderm located dorsal to the sickle-shaped blastopore.

The blastoporal graft adheres firmly to the non-coated endoderm and soon sinks into the latter to form a flush surface with it (fig. 26 b). However, complete invagination is not achieved, evidently because many of the prospective chorda-mesoderm cells are converted regulatively into epidermis and form a peripheral cap which acts like the epidermis in the preceding experiments, preventing the mesodermal tissues from moving into the center of the endoderm.

Sections (17 cases) show that in accordance with the results of previous workers (reviewed by Holtfreter, '38), the grafted blastoporal material produced regulatively not only epidermis but also some neural tissue and pigment cells. If the graft contained prospective foregut endoderm, this coated material did not invaginate into the uncoated trunk endoderm but spread over the latter and competed with the epidermis to form an epithelial covering over the mesoderm. The mesodermal differentiations consisted of notochord, somites, mesenchyme and pronephric tubules which sometimes grew out into blunt projections (fig. 26 d). Antero-posterior elongation and bilateral disposition of the axial tissues were not as perfect as might have been expected.

It is well known that if such blastoporal material is grafted into or underneath the ventral ectoderm of a whole gastrula, it invaginates and establishes a complete axial system, partly out of its own substance, partly by way of recruiting into this system induced host cells (Spemann and Mangold, '24;

Mayer, '35, et al.). Even explanted blastoporal primordia undergo complicated re-arrangements of their cells and then segregate into axial tissues which tend to exhibit a bilaterally symmetric and antero-posterior configuration (Holtfreter, '38). This occurred only if the explants became attached to a surface, such as glass; non-attached explants remained balled-up and failed to achieve a typical axial tissue arrangement. The occurrence of similar irregularities in the present preparations may be ascribed mainly to the fact that the epidermal-endodermal covering was both too fragmentary and unsuited to allow for an axial arrangement of the chordamesoderm.

4. Recombination of dissociated cells from several dorsal blastoporal lips. The free cells from several blastoporal explants — 6 in most cases — were mixed and recombined. Preparations of this kind have been made and briefly discussed by Holtfreter ('43c).

Due to the presence of cephalic endoderm and the formation of epidermis by way of regulation, the cells determined to become mesodermal move away from the surface of the initially globular re-aggregates. This process of sorting-out cannot be followed histologically as closely as was possible with the more distinctly characterised cells from neurulae. It is quite likely that the transformation of prospective mesoderm into epidermis occurs only at the periphery of the explant, not involving any centrifugal cell movements.

What can be observed in sections (22 cases) is this: After 48 hours the surface of the re-aggregates consists of a mosaic of endodermal and epidermal tissue patches. Subsequently, the epidermal patches, having a greater spreading tendency than the endodermal cells, tend to fuse into larger patches; but not enough epidermis is formed to cover the whole explant. It need hardly be mentioned that the peripheral fore-gut epithelium has its ciliated or otherwise active surface turned outside, such as has been observed in many corresponding situations, e.g. in exogastrulae.

The mesodermal tissues collect as a single mass underneath the peripheral endodermal-epidermal epithelium and then differentiate into the same kind of cells as were listed in the preceding experiment on non-dissociated material. Of special interest is the fact that the notochord usually occurred not in the form of many scattered bits, but as a single mass which however failed to exhibit pronounced elongation, and possessed abnormal branchings. Notochord was as a rule associated with quite typically segmented somites whose myoblasts elongated tangentially to the notochord. The regulatively formed neural tissue also tended to appear associated with notochord; it became subject to the molding influences of both notochord and somites which have been pointed out by previous workers (see Lehmann, '45). The pronephric tubules appeared preferentially next to mesothelium-lined (coelomic) cavities which were scattered throughout the explant. One major continuous pronephric tubule always developed and occasionally one or two small accessory pronephric islets were found. As compared with the corresponding non-disaggregated material the degree of tissue reorganization was somewhat less perfect: no indications of antero-posterior or bilateral axiality were noticeable. Considering the fact that these preparations consisted of the cell material from several, thoroughly mixed blastoporal isolates, the reorganizations obtained are remarkable enough to deserve further inquiries.

In an attempt to interpret the results, several factors must be taken into account. The two organizing factors which became so clearly revealed in the preceding experiments with older cell material, namely directed locomotion of cytologically determined cells and selective cell adhesions, were obviously also operative here. The least that can be attributed to them is the segregation between endodermal and mesodermal tissues.

The question of how the tissues within the prospective mesoderm became segregated leads to the problem of embryonic fields. Numerous workers have found that the prospective chorda-mesoderm of the early gastrula is not locally determined but has the regulative capacities of a field. The present data suggest that the cells from several blastoporal lips, when mixed tend to establish a new, supranormal, and yet integrated embryonic field. The mixed and re-aggregated cells of 6 such fields, instead of forming 6 notochords and 12 pronephric tubules, form much less of these tissue units, usually a single though abnormally shaped notochord. In view of these results and those of other workers we may conclude that inductive cell determinations occurring amongst the components of the centrally accumulated mass of mesoderm have played an equally important role in the process of tissue-patterning as did purely cell kinetic processes. The latter are one of the consequences of inductive cell transformations.

General considerations and conclusions

Two general conclusions emerge from the foregoing experiments. (1) The different cell types of the amphibian embryo, whether present as single cells, cell sheets, or globular cell masses, exhibit tissue-specific tendencies of moving either centrifugally or centripetally within a composite cell aggregate. (2) Directed movements are followed by the phenomenon of cell-specificity of adhesion. The combined effects of these processes necessarily result in segregations and recombinations of tissue primordia or individual cells.

Analysis is difficult if the cell types cannot be clearly distinguished from each other, or if in the course of experimentation the cells enter into new pathways of differentiation. In other words, such experiments must be carried out with well defined types of cells that are no longer responsive, or not exposed to inductive or other determinative influences of their new environment.

Difficulties of this kind turned up in the experiments on mechanically dissociated and re-associated cells of sponges (Wilson, '08; Galtsoff, '25; Brondsted, '36, et al). There, as in our experiments, the re-aggregates formed typical tissue patterns, but the principles of directed cell movements and cellular transformation could not be clearly defined as separate agencies to bring about the organismic reconstitutions.

We have encountered similar complications in the experiments with the cytologically undetermined gastrula material. But in most of our experiments the possible interference of induction was avoided by using more definitely determined material from older stages and by dealing with cell types which, according to their germ layer derivation, are clearly distinguishable both in living and sectioned preparations. This made it possible to trace exactly the locomotions and subsequent differentiations of the cells employed in the various combinations. Let us examine in more detail the different aspects of the problems raised here.

Unsupported hypotheses on morphogenesis

Many of the data obtained from this study do not support some of the hypotheses of earlier workers regarding the mechanisms of gastrulation-neurulation. There are three hypotheses which warrant consideration in this regard.

1. The notion of His (1874) that embryonic infoldings or outfoldings are caused by locally increased cell proliferations has already been refuted by numerous investigations on the mitotic activity of embryonic areas in various animals (see Holtfreter, '43b; Gillette, '44). The extensive studies on localised vital staining further invalidate this notion. In agreement with previous reasonings, this argument may be added: the invaginations or spreadings in normal or experimental tissue combinations occur within 10–15 hours (amphibians), when hardly any cell divisions and certainly no mass increment occur. Clearly, the principle of growth is no more involved in these processes than it is in the filling of a sports arena by people. It is an anachronistic trend of many embryologists to speak of ingrowth, outgrowth or overgrowth when referring to gastrulation movements.

2. Other speculative minds ascribed infoldings to imaginary pushing forces of the surrounding epithelium (Bütschli, '15; Giersberg, '24) or to localized water intake (Glaser, '14, '16; Spek, '31). The findings of subsequent investigators (Brown, Hamburger and Schmitt, '41; Holtfreter, '44) as well as the results of this study provide no factual evidence for these suppositions.

3. More recently, Lewis ('47, '49) has interpreted the gastrulation-neurulation processes as resulting from localized contractions of a rather ill-defined superficial gel-layer. This concept cannot be reconciled with the experimental data of Holtfreter ('44, '48), Devillers ('48, '50) and Trinkaus ('49, '51). The present experiments furnish numerous additional data which disqualify the assertions of Lewis. They demonstrate that the inward movement of one cell type underneath or into a layer of other cells has nothing to do with the contraction of a superficial gel-layer. This layer can hardly be interpreted as anything else but the coat.

It is true that, under normal conditions, the coat is in a state of elastic tension, comparable to that of an expanded rubber film. The coat does contract when released from tension, for example when fragments of neural plate or blastoporal region are isolated. The fragments immediately curl into concave bodies (Holtfreter, '43a, b). But due to the antagonistic forces of lateral cell adhesion, such precociously involuted fragments soon flatten out again. It is inconceivable that epiboly or the formation of such deep cavities as the archenteron could result from curlings due to the contraction of any kind of surface structure. At any rate, single cells, or uncoated cell masses, move into the depth just as well as does a layer of coated cells.

In the areas of medullary plate or blastoporal region, the coat unites the invaginating cells peripherally but it does not provide the motive power for invagination. On the contrary, the coat counteracts invagination and favors (in an apparently paradoxical way) epibolic spreading. This is strikingly illustrated by the fact that a piece of epidermis which has been deprived of its upper coated cell layer, will slip underneath the surrounding coated epidermis or into a mass of denuded endoderm (Holtfreter, '43a, '44). Similarly, as shown here and in previous experiments, coated endoderm tends to spread over uncoated endoderm, or mesoderm, whereas uncoated endoderm or mesoderm will invaginate into uncoated endoderm. Hence the initial decrease of surface area (contraction of the coat), as observed in the infoldings of medullary plate, blastoporal region, etc., is the result and not the cause of invagination. These and many other data (see below) point to the conclusion that invagination as well as epithelial spreading result from an active locomotion of cells or tissues and that the coat merely assists in integrating these collective cell movements.

Dalcq and Pasteels (Dalcq, '41) have postulated that invagination of the blastoporal material is caused by an interplay of two hypothetical concentration gradients. It would require much imagination to find support of this hypothesis in the present results.

Classification of cell movements. In his fundamental analysis of the morphogenetic cell movements of the amphibian embryo, Vogt ('29) distinguished between epiboly, invagination, convergence, divergence and axial elongation. Our experiments have contributed little to an elucidation of the last three phenomena. On the other hand, they indicate the presence of another tendency, namely that of an outward movement of cells which is different from evagination and which may be designated as emigration. Epiboly which is normally associated with the coated ectoderm, can be exhibited also by coated endoderm; so the less committal term "spreading" will be applied to this common phenomenon.

To designate inward movements, a diversity of words have been used in the literature: invagination, involution, infolding, immigration, invasion, ingression, and ingrowth. These terms come in handy for descriptive purposes but they have no explanatory value. Evidence shows that the cells of one and the same primordium can, under different experimental conditions, display any one of the processes designated by the above terms (except ingrowth). It seems justified therefore to make no special efforts to distinguish between the various terms and speak of invagination when referring to any kind of inward shiftings.

Tissue-specificity of morphogenetic movements

The present experiments were aimed at studying both the cell-inherent kinetic tendencies of embryonic primordia and the environmental conditions which favor or impede their expression. That the phenomena of spreading or of axial elongation and segregation are actually due to tissue-inherent tendencies has been shown previously by culturing isolated pieces of ectoderm, endoderm, or blastoporal lip on an unspecific surface, for example on glass (Holtfreter, '38, '44). Without an organic or inorganic surface to adhere to, the primordia cannot spread, and axial stretching remains abortive. Nor can invagination occur in the absence of an embedding matrix of other cells. The present data throw some light upon the question of why the primordia in an intact embryo move about and segregate as they do normally.

In the following discussion the kinetic manifestations of the different primordia will be treated separately.

(a) Prospective coated epidermis of gastrula or neurula stages never moved into the depth of other tissues (inductions excluded). Instead, it spread over mesoderm, uncoated endoderm or medullary tissue. When competing with coated endoderm, the ectoderm exhibited a relatively stronger spreading tendency than the endoderm. When individual ectodermal cells were interspersed in between either endodermal, mesodermal or medullary plate cells, they moved to the surface of the aggregate and then established, by way of spreading, a continuous surface epithelium.

Centrifugal emigration of scattered cells, or of nucleated cytoplasm, to a peripheral position is a familiar phenomenon in arthropod eggs. In a less conspicuous way it occurs also in normal amphibian embryos, namely when the multilayered ectoderm changes into a thinned-out epidermis (Holtfreter, '43b; Gillette, '44). After the initially buried epidermal cells have attained a surface position, they secrete a coat and it appears that, concomitantly, they acquire a pronounced tendency for epibolic spreading.

(b) Endoderm. It seems strange at first sight that the endoderm cells also have the tendency of centrifugal locomotion and surface spreading. This tendency is manifested in preparations which consist only of endoderm and mesoderm but it is inhibited when the surface is already occupied by ectoderm. Coated endoderm has a definitely stronger spreading tendency than uncoated endoderm. The experimental data appear less strange in view of the fact that in normal development endodermal expansions also occur, viz. in the dorsad movement of the walls of the archenteron and in the subsequent increase of surface area in the elongating gut. In whole embryos it is the relatively stronger spreading tendency of the ectoderm which prevents the endoderm from expanding into a peripheral epithelium.

This implies that an internal archenteron cannot be established, and not even be maintained, in the absence of an epidermal covering. Inhibition of ectodermal epiboly produces exogastrulae whose endoderm forms the outer envelope of the mesoderm. Removal of the whole ectoderm at neurula or even older stages causes a complete inversion of the remaining two germ layers, largely brought about by an outward movement and spreading of the coated endoderm of the archenteron over the denuded surface of the embryo (p. 81). If a wound is inflicted on an advanced embryo, reaching into the archenteron, the coated epithelium of the latter moves outward through the hole until it meets the epidermis. Since coated endoderm and epidermis cannot spread over each other, such a fistula remains forever open.

Invagination of the blastopore. Although the above as well as previous data demonstrate that the spreading of coated endoderm is counteracted by the relatively stronger spreading power of coated ectoderm, it follows by no means that invagination of the blastoporal endoderm results from a pushing action of the expanding ectoderm of the gastrula. In the complete absence of this ectoderm, a fragment of the blastoporal lip will move into a socket of uncoated endoderm, although the coated endodermal portion of the graft tends to spread over the socket (p. 89). But a ball of uncoated endoderm moves entirely into a layer of uncoated endoderm. This alternative exhibition of the opposite movements of spreading and invagination by the endoderm requires an explanation. The following one may be offered.

In blastoporal invagination, the factors making for invagination must evidently overcome those making for peripheral spreading of the endoderm. Since spreading tendency can be associated with the presence of a coat, and invagination with uncoated cell surfaces, it seems that the endoderm of the blastoporal lip moves inward due to a response of the inner, motile cell surfaces to some "cytotactic" or other directional influence of the inner milieu. The coated outer surface yields passively to the pulling force of the actually observed inward movement of the cells. Soon after invagination has begun, the adjacent marginal mesoderm becomes involved; its pronounced invagination tendency no doubt contributes considerably to the completion of the formation of the archenteron.

Evaginations. No special endodermal differentiations were recorded because in most of these experiments the bulky endoderm of the prospective trunk region was used. Even in normal embryos, this material fails to attain differentiation and degenerates eventually (Holtfreter, '33). The primordia of foregut or stomach which did differentiate in mesoderm-endoderm aggregates, formed a peripheral epithelium, the secreting or ciliated side of which was turned outside. If gastric glands developed in this everted epithelium, they would sink into the underlying mesenchyme. To follow the time-honored terminology, here one must speak of an invagination and not, as in normal development, of an evagination of the glands. (For illustrations of this phenomenon, see Holtfreter, '33, '38).

The data indicate that invagination and evagination are actually controlled by the same principles and that these two terms merely refer to topographic relationships and not to different mechanisms. Both the ecto- and endoderm are surface epithelia which protect the mesoderm from the vicissitudes of the external medium. The tubular glands which develop from both kinds of epithelia represent in either case shiftings away from the extra-organismic medium.

(c) Medullary plate. In combination with either prospective epidermis, endoderm or both, fragments or individual cells of the medullary plate invariably moved into the core of the aggregate where they formed a homogeneous neural tissue. The same occurred when mesenchyme from the neural fold or from the mesoderm mantle was added to the above combinations.

It is obvious that the tendency for inward movement is present in each medullary cell and that to manifest it, a matrix as atypical as a mass of epidermal or endodermal cells is sufficient. Hence neither the surface coat nor the expanding epidermis, or the mesodermal substratum can be made specifically responsible for the infolding of the medullary plate in whole embryos. We hasten to add, however, that the coat and the various embedding tissues do play an important role in normal embryogenesis, namely in the subsequent shaping of the invaginated medullary material.

Some of these environmental (not "inductive") influences are concerned with the formation of a characteristically shaped *central lumen of the neural tube.* In the invaginating whole neural plate, or in fragments of it, the contractile coat remains intact. Although not causing invagination, the coat nevertheless contracts, retains its distal non-adhesiveness and becomes significantly engaged in the formation and maintenance of the central lumen.

The above data show that central lumina can be formed in the absence of a continuous coat, namely by way of secondary cavitation within a compact mass of prospective neural tissue. Subsurface formation of a neurocoel has been observed before in experiments in which the ectodermal layer was too thick to become entirely neuralized by the underlying inductors. This type of secondary neurocoel formation is the norm in teleost development. Under the term of polymyely, similar secondary cavitations have been described in experimentally injured brains and in tissue cultures.

Evidently, secondary lumen formation is caused by the exudation of a liquid which increasingly separates the initially contacting cells. Localized liquid formation presupposes an alignment of axially organized cells around their future lumen. Such a preparatory cell arrangement could actually be observed in our aggregates containing centrally accumulated medullary tissue. The difference between lumen formation through infolding and through secondary cavitation is not as great as it appears if one considers the fact that the infolded lumina are likewise kept open thanks to a continuous secretion of a liquid from the walls of the lumen.

Prospective neural tissue, like epidermis and gut epithelium, seems to be capable of transferring water from the outside to a central lumen. In epidermis-covered explants of neural tissue this process may result in an excessive blowing-up of the neurocoel cavities (p. 68). The opposite occurred in preparations in which the invaginated medullary cells became covered by endoderm and lacked a mesenchymatic mantle. Here the infolded lumen disappeared and no new lumina developed by way of cavitation (pp. 70, 76). In a still undisclosed way, a purely endodermal environment seems to prevent the transfer of water to the medullary tissue. Normal embryogenesis does not have to cope with this situation.

Lumen formation through infolding versus secondary cavitation or delamination. There is the well-known phenomenon that, varying with the taxonomic groups, the same kind of internal cavity can be established either by way of infolding or through secondary cavitation. Examples: archenteron, coelom, neural tube, lens, otocyst, and various structures in invertebrates. These alternative modes of lumen formation are distributed so erratically amongst the taxonomic types that they cannot serve as a basis for establishing phylogenetic relationships. The present data underline the insignificance of the different kinetic devices used to achieve the same result. One and the same primordium, such as the cells of the medullary plate, can resort to the normally not employed device of secondary cavitation if they attain an internal position without infolding. Similarly, in certain instances of tumor growth, e.g. in carcinoma of the mammary gland, follicles are produced through secondary cavitation rather than from outgrowth of the duct system as occurs normally.

It appears that, in a general sense, internal epithelial lumina can be established or retained only if there is a gradient of some sort between the intra- and extraorganismic milieu. The medullary plate of an embryo remains a flat layer (spina bifida) in the absence of an epidermal covering. On the other hand, a piece of prospective epidermis which is grafted into the mesenchyme, or into the coelom of older amphibian larvae, either loses its coat and disintegrates into scattered cells or it forms, by way of rounding-up and secretion, a vesicle, whose coated surface is turned inside as in all invaginated epithelia (Harris, '42). Likewise, the formation of an internal gut lumen presupposes the existence of another epithelium (epidermis) which shields the endoderm from the external medium. Therefore, the decision of ectoderm or endoderm to form either an external layer or an internal cavity depends upon the topographic relationship of these epithelia to the external medium. None of the germ layers can form tubes or cavities in the absence of enveloping tissues.

Regional organisation of the neural system. This problem is far too complex and the data we can offer to solve it are too fragmentary to justify a lengthy discussion. It may be mentioned, however, that the present results confirm the often stated notion that the shape of the neurocoel and the bilateral and antero-posterior organization of the neural system are largely dependent upon continued formative influences of the various tissues of the archenteron roof. Consequently, little regional organization was obtained in combinations in which the tissues of the archenteron roof were absent or had been deranged together with those of the medullary plate. The multiplicity of atypically shaped neurocoels as observed in these combinations would probably not have occurred if the disarranged and re-associated cells of the medullary plate had been subsequently exposed to the formative influences of a normal archenteron roof. Nevertheless the method of reshuffling the cells of a disaggregated primordium represents a new and promising approach to investigating the baffling problem of embryonic fields. A diversity of experiments along these lines are in progress.

(d) Mesodermal primordia. In any of the combinations employed, cells determined to form mesodermal tissues moved into internal sites irrespective of whether the embedding material consisted of ectoderm or endoderm and of whether the mesoderm represented scattered cells or solid cell masses. The mesectoderm from the neural crest behaved similarly. However, the final allocation of the sorted-out mesoderm varied with the kind of tissue association. If combined only with endoderm or ectoderm, the mesoderm took up a *central position;* but if ectoderm cells were added to the endoderm, or if medullary cells were added to the ectoderm, the mesodermal cells occupied an *intermediary position* between the centrally accumulating endoderm, or neural tissue, and the peripherally established epidermal layer. To attain this intermediary position in an initially mixed-up cell aggregate, the centrally dispersed mesoderm cells must have moved centrifugally and the peripherally scattered mesoderm cells must have moved centripetally.

Evidently the direction of movement of the mesoderm in these combinations is controlled by the presence or absence of an epidermal covering. The behaviour of the mesoderm is only partly similar to that of the endoderm. The cells from both germ layers tend to invaginate in the presence of ectoderm, but when endoderm alone is combined with mesoderm, the latter invariably moves into the former indicating a comparatively stronger invagination tendency and the absence of a spreading tendency within the mesodermal primordia.

With reference to normal embryogenesis, these data show that the formation of a mesoderm mantle over an endodermal core requires the presence of an epidermal covering. Even after it has attained its intermediary allocation, the mesoderm cannot maintain it without a protective epidermal covering. When the latter is removed in advanced stages, the whole mesoderm mantle breaks up into patches which then move into the underlying endoderm (p. 81).

Investigation of the factors involved in the *regional and bilateral organization* of the mesodermal primordia transgresses the scope of this paper. Let us mention however that although the mesoderm can achieve tissue segregation and histotypical differentiation inside an endodermal envelope, it fails under these conditions to exhibit typical caudal stretching and bilateral arrangement of its components. Evidently such an envelope cannot substitute for the epidermis to allow for a manifestation of the inherent tendencies of the mesoderm to establish a typically organized axial system.

The intermingled cells of disaggregated blastoporal lips showed a more pronounced capability of regulating into an axial system than did those of the archenteron roof. This agrees with the current notion that the parts of the mesoderm become more definitely determined during the period of gastrulation. To state it differently: whereas the tissue segregations within the mesoderm, as observed in the experiments with neurula and older stages, can mainly, if not entirely, be attributed to directed cell movements, the more perfect reconstitutions as obtained in disarranged blastoporal lips are at least partly due to inductive interactions amongst the parts of the lips, each lip representing initially an "embryonic field." It should be pointed out in this connection that histological regulations or reconstitutions of any kind probably always involve considerable shiftings of cytoplasm or cells resulting in the reconstitution of a new organismic field.

Factors determining morphogenetic cell movements

The above data, as stated in histological terms, raise problems on cellular and physicochemical levels. Unfortunately, no satisfactory answer can be given to the question of what makes the cells move either inward or outward under normal or experimental conditions. As a basis for further investigations let us point out the crucial phenomena and the problems that arise from them.

The *tendency for directed cell movements* has been shown to be tissue-specific whereas the environmental factors enabling the execution of these movements are far from specific. Consider the spreading tendency. The literature on tissue cultures provides ample evidence that a spreading of all sorts of epithelia can take place on such atypical surfaces as glass or clotted blood plasma. Under these conditions, epithelia like those of the gut, of endodermal glands, or of the kidney, will form membranes rather than tubular cavities. These epithelia may exhibit inward movement and tubular configuration in the presence of a substratum which need not be specific but simply allows the cells to move away from the interfacial conditions between substratum and external environment. Examples: medullary tissue, or mesoderm, migrating into such atypical substrata as endoderm or ectoderm, or explanted kidney epithelium moving into the fibrin matrix of a clot of blood plasma (Rienhoff, '22) to form tubules. Grobstein ('53) observed in the case of salivary glands from mouse embryos that the type of mesenchymatic matrix influences the extent of tubular ramification of these glands.

While acknowledging the enhancing and specifying influences of the tissue environment upon shape and perfection of any invaginated epithelial structures, there remains the more general problem of why certain tissues tend to spread or to invaginate at all. What does cellular kinetic "tendency" mean in a more concrete sense?

The *studies on isolated cells* provide some important though not sufficient clues for an understanding of morphogenetic mass movements. There is no doubt that the isolated cells of the different germ layers here investigated do exhibit tissue-specific locomotion and that their direction of locomotion is controlled by an inherent cellular axial polarity. For instance, singly isolated cells of prospective epidermis flatten and spread on a contact surface while those from the medullary plate or the bastoporal lip do the opposite: they elongate along their proximo-distal axis and migrate, with their non-coated inner cell pole leading the way, a tendency which normally leads to invagination (Holtfreter, '47a, b).

The ectodermal primordia of neural system, lens, nose, or ear vesicle, *acquire the tendency to invaginate as a consequence of induction.* In the case of the medullary plate, it could be shown that the cylindrical elongation of the constituent cells, as seen in sections, is due to a newly arising stretching tendency of each cell and not, as has been suggested by Schmitt ('41), to contact influences between neighboring cells (Holtfreter, '47). Evidently, induction involves the elaboration of structural properties of the cell membrane which are absent in pre-induction stages and which cause the induced cells to elongate and to invaginate instead of spreading as does the non-induced epidermal portion of the ectoderm. It should be mentioned, however, that cell elongation and invagination tendency are not necessarily linked up with induction. For instance the blastoporal endomesoderm exhibits these phenomena in the absence of adjacent inductors.

Amoeboid motility in any cell type of the amphibian embryo is produced by an axially organized succession of alternating expansions and contractions of the cell surface which need not be associated with gel-sol transformations of the inner cytoplasm (Holtfreter, '47a, b). If, as in isolated epidermal cells, this three-steps-forward, two-steps-backward succession of movements is executed by the entire peripheral margin of the attached cell, the net result is centrifugal spreading of the cell. If the extension movements are confined to only the "anterior" part of the cell, as in medullary and other non-spreading cells, they cause unidirectional locomotion or, if the cell body remains sessile, the projection of localized, sometimes highly extended amoeboid processes, as in a neurone.

To what extent can the observations on isolated cells further an understanding of the mechanisms operating in the *directed cell movements in* complex tissue combinations? Definitely more than has been conceded by Vogt ('13), Kuhl ('37) or Bonner ('52).

One can interpret ectodermal epiboly as due to the summative effect of the spreading tendency of each of the constituent cells of the cell sheet. In gastrulation and wound healing, the extent of epidermal spreading seems to be simply a matter of readily available, non-coated cell surfaces which controls the manifestation of the ever present spreading tendency of the individual cells.

Difficulties of interpretation arise in the case of invagination movements. To be sure, the penetration of the blastoporal endoderm into the inner endodermal cell mass of the gastrula seems to involve the same principles which direct the locomotion of the isolated cells from this region. The interdigitations between blastoporal cells and the inner endoderm support this view. Also, the immigration of individual cells of mesoderm or neural plate into a non-specific cellular matrix can be largely attributed to the mechanism of amoeboid locomotion. Most likely, this inward movement of scattered cells into a densely packed layer of other cells proceeds without the formation of hyaline pseudopodia and of gel-sol transformations.

The above interpretations fail however to explain the inward movement of large cell masses. The neural plate, a piece of forebrain, or a lump of mesoderm consisting of hundreds of cells, glide into the depth of a cellular matrix with a perfectly smooth periphery. This excludes intervention of the extension-contraction mechanism of the cell membrane which makes for the locomotion of isolated cells. And yet, compact cell masses showing no amoeboid locomotion of their peripheral cells, move inward just as readily as do individual cells.

The riddle of this situation comes somewhat closer to a solution if it is realized that the shifting of a tissue into or underneath another tissue always involves reciprocal accommodating cell movements of the latter. When a lump of neural or mesoderm cells is placed upon the smooth surface of endoderm, the loosely connected cells of the endoderm actively move around the graft, comparable to the phagocytic movement of an amoeba around a food particle. It appears that while in some instances the amoeboid movements of both the invading and embedding cells are equally instrumental in bringing about inward shiftings, in other instances it is mainly or entirely the engulfing tendency of the embedding cells which achieves the same end-result.

These considerations do not tell us why, in embryos or in composite aggregates, certain cell types move inward and others outward. It seems necessary to assume the *existence of a concentration gradient of some sort between inner and outer milieu of the aggregate towards which the different cell types react differently.* To indicate the general nature of this problem, the experiments of Roudabush ('33) may be mentioned. They showed that when a Hydra is turned inside-out, a normal Hydra can be reconstituted due to an "amoeboid" inward movement of the everted endodermal cells and an outward movement of the ectodermal cells. Probably similar directed cell migrations are involved in the reconstitution of dissociated sponges.

By way of analogy one may compare invagination with amoeboid phagocytosis or with the incorporation of an oil drop of a relatively high surface tension into the denuded egg of a sea urchin having a low surface tension (Kopac and Chambers, '36). Similarly, epibolic spread might be interpreted by assuming that the interfacial tension between coated epithelia and the ambient aqueous medium is lower than that between uncoated cells and this medium. Rhumbler

(1899, '02, '27) and Holtfreter ('43b) have elaborated upon such speculations. They suggested that if the blastocoel, or the interior of a cell aggregate in general, contains surface tension-lowering substances, invagination may be due to a kind of cytotactic reaction of the proximal cell surfaces to a gradient of interfacial tension between inside and outside of the embryo. The cells or cell masses would move toward the surface tension-lowering gradient comparable to the engulfing of certain kinds of oil drops by a sea urchin egg.

As has been emphasized by Rhumbler (op. cit.), the forces of molecular interfacial tension cannot be ruled out in the kinetics of embryonic cells whose surface is of a semiliquid nature. Yet the well-attested structural properties of the cell membrane would necessarily resist a free display of these forces since the latter presuppose freely mobile molecules. Further work is required to clarify the significance of interfacial tension in morphogenesis and to define the factors which direct the tissue-specific movements here recorded.

Cellular adhesiveness

The phenomenon of cellular adhesion is the prerequisite for the evolution and ontogenesis of multicellular organisms. In early cleavage stages this phenomenon may be almost absent, as in the eggs of mammals, or echinoderms, where the blastomeres round-up and retain hardly any contact with each other. Their falling apart is largely prevented by the tightly applied hyaline layer (echinoderms) or by the egg membranes. In the case of the amphibian egg, the syncytial coat is at least as important as is intercellular adhesion to hold the blastomeres together. In all these instances the function of the external coverings as a mechanism to insure cohesion of the blastomeres is progressively replaced by an increasing adhesiveness of the adjacent cell walls. At the gastrula stage, the cells of the peripheral layers have attained mutual adhesion, whereas the cells of the inner endoderm are still lacking it.

In any of the above recombinations, the different cell types at first adhered to each other irrespective of germ layer derivation and of differences of species. The cells tended to form a tightly packed globular aggregate which reflects their tendency to establish maximal mutual contact and to reduce their interface with the ambient aqueous medium to a minimum. However, this initial adhesiveness is static only if the aggregate consists of one kind of cells. In heterologous combinations, subsequent sorting-out movements lead to new cell arrangements. This suggests that the molecular bonds making for indiscriminate cell adhesions are somewhat unstable.

The question of an extracellular cementing matrix. There is no evidence that embryonic amphibian cells migrating on glass secrete a sticky substance that facilitates locomotion as in the case of the amoebae (Jennings, '04) and amoebocytes of invertebrates (Fauré-Fremiet, '29). There is no indication either that Weiss' ('45) observations on tissues cultured in clotted blood plasma are pertinent with regards to cellular movements and adhesions in early embryos because the latter do not seem to possess extracellular matrices comparable to a fibrin coagulate.

The question of whether or not the embryonic cells adhere to each other with their "naked" surfaces or by means of an intercellular cement appears to be largely a matter of definition. The presence of such a cement seems to be well documented in the case of various epithelia or endothelia in adult organisms (Gray, '26; Chambers, '40). Clearly, the peripheral coat represents an exudation product of this kind; but the coat is non-adhesive at its outer surface and it does not extend down to the cellular interfaces which appear to touch each other directly (Holtfreter, '43a; Dollander, '51). Nor does the free mobility of isolated cells along each other's surfaces indicate the intervention of an adhesive cement (Holtfreter, '47a). On the other hand, it has been observed that the same external conditions which cause dissipation of the coat, also destroy mutual adhesiveness of the non-coated cells. This suggests a constitutional similarity between the well-defined coat and the microscopically non-detectable cement which holds the inner cells together.

It should be pointed out that whereas the cell-disaggregating effect of alkali, citrate or oxalate is reversible, that of proteases is irreversible (Townes, '53). This shows that calcium ions and proteins are involved in adhesion, but it is a matter of taste to consider the proteolytically susceptible material as an intercellular cement or as part of the living cell membrane.

Morphogenetic movements and cellular adhesiveness obey quite different controlling factors although both result from specific properties of the cell membrane. Prospective neural tissue invaginates while losing adhesion to ectoderm or endoderm, whereas mesoderm invaginates while retaining adhesiveness to the embedding tissues. Not only in the latter instance, but in cellular locomotion in general, the crux of the matter is that a cell migrates in spite of being adhesive to an organic or inorganic surface of contact. In morphogenesis, the forces controlling directed movements must overcome those of cell adhesion. It becomes evident once more that the molecular bonds involved in these early, temporary and indiscriminate cell adhesions are extremely labile and as such not comparable to the bonds involved in antigen-antibody reactions.

Tissue- and stage-specificity of cell adhesion. The situation becomes more involved because of two sets of experimental data: (1) The indiscriminate adhesiveness of early embryonic cells changes subsequently into tissue-specific selective adhesiveness; (2) this change cannot readily be correlated with the stages of cellular differentiation.

As to the first point, the present as well as previous data show that initially intermingled different cell types, after they have undergone sorting-out by way of directed movements, stick together permanently to form homogeneous tissues which segregate themselves from adjacent, other types of tissues. Whether this segregation results from a process of "pinching off" or through cleft formation seems to be irrelevant. One and the same tissue may follow either of these procedures according to the environmental conditions; for example, prospective neural tissue separates from epidermis or endoderm under any circumstances — obviously due to an arising disaffinity between these tissues. On the other hand mesodermal tissues retain a permanent adhesiveness to ecto- and endodermal tissues. New cellular affinities arise in the course of development (Holtfreter, '39).

As to point two, the above experiments dealing with the combination of primordia of different developmental stages (p. 73) show that the alternative between indiscriminate and selective adhesiveness is not simply determined by developmental stage. A piece of forebrain, or of mesoderm, which has already undergone invagination and segregation will, under experimental conditions, once more adhere to a layer of early ectoderm or endoderm and move into these layers. Invagination and subsequent segregation of these tissues of different age require about the same time interval as is observed in corresponding tissue combinations of an earlier stage. Hence it appears that the factors making for indiscriminate and selective cell adhesion may be present simultaneously and that it requires a prolonged contact between heterologous tissues to bring forth the selective component of adhesion.

There is another point worth mentioning though it is partly implicit in the above statements. It is the fact that adhesiveness between different tissues may vary in *degree*. Dissection of a fresh neurula shows for instance that the prechordal mesoderm is much less firmly attached to the medullary plate than is the notochordal tissue. Alkali treatment of advanced embryonic stages does not produce the ready cell dissociations as it does in earlier stages.

Hypotheses concerning the molecular factors engaged in cellular adhesion. The hypothesis of Weiss ('41, '47, '50) which interprets selective cell adhesion in terms of an interlocking of complex molecules possessing complementary configurations is based on Pauling's theory that the strength of a chemical bond is proportional to the complementariness or fit of the molecules. Weiss has diagrammatically portrayed some of the possible molecular configurations that might account for selective cell adhesion. While it should be realized that no precise data as to the configuration of the cell-binding elements are available, Weiss' hypothesis sounds persuasive. Nevertheless, the non-static nature of cell adhesions, as pointed out above, leaves room for other hypotheses envisaging a more complex and less rigid lock and key mechanism.

In a more specific way, Tyler ('46) attempted to explain cell affinity in terms of an antigen-antibody complex. He proposed that the molecules at the cell surface react with molecules of adjacent cell surfaces in a manner analogous to the union of an antigen with its antibody. However, according to the more generally accepted concepts of immunological reactions, combinations of this type would not appropriately be considered as antigen-antibody complexes. Obviously in the present case, no foreign element is introduced which could elicit formation of an antibody, a condition which is essential for an antigen-antibody reaction. The auto-antibody concept, by not sharing this basic requirement, may be applied to describe the combination or chemical interaction of any complex molecular species, not only linkages between interfacial cell surfaces, but innumerable others even as remote as the recombination of dissociated hemoglobins. Under these circumstances, referring to the molecular species responsible for cell adhesion as antigens yields little clarification of the essential problem.

In view of the rather complicated nature of the phenomena observed and of our ignorance concerning the physico-chemical factors involved, one could think of a diversity of symbolistic schemes which might tentatively cover the actual situation without, however, explaining it. One of the possible interpretive illustrations which suggests itself is that of figure 27. In these diagrams it is attempted to combine the changes of adhesiveness with the *simultaneously* occurring directed cell movements, as examplified in the combination of prospective epidermal and neural cells. Only a few words need be added to these self-explanatory diagrams.

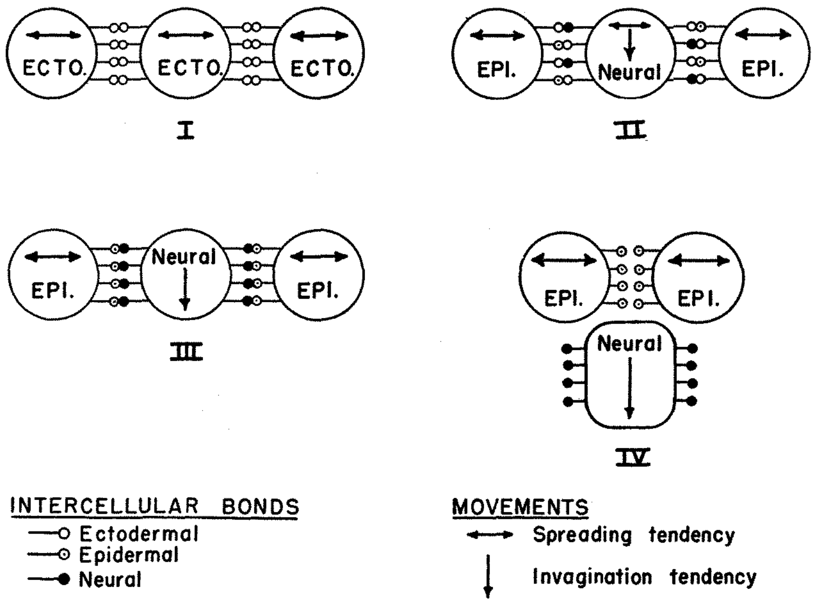


Fig. 27 Schematic representation of changes in intercellular bonds and migration tendency leading to segregation of medullary and epidermal cells (or tissues). Stage I. Early condition of ectodermal cells having a moderate spreading tendency and non-specific intercellular bonds. Stage II. Partial conversion of ectodermal into epidermal and neural cell types, indicated by changes of the unspecific bonds into neural and epidermal types. Stage III. Completion of bond modifications and emergence of cell-specific migration tendencies. Stage IV. Invagination and segregation result from these cellular modifications. In artificial aggregates of neural and epidermal cells, the cells may be considered to have been returned to the condition of stage III from which stage IV is once again attained.

While the cells of different prospective fate either spread or invaginate due to cell-specific migration tendencies which are evidently directed by inside-outside gradients within the aggregate, their mutual adhesiveness is only temporary and not very firm. Hence the prospective neural cell of figure 27 can readily slip into the depth whereas the prospective epidermal cells continue spreading (indicated in arrows). By the use of different symbols for the bonds of cellular adhesion it is indicated that the originally indiscriminate adhesiveness of the early ectoderm (stage I) changes into cell-specific adhesiveness of respectively epidermal and neural cells (stages II and III), making for a separation of the two cell types or tissues.

While the experiments reported clearly indicate that changes in adhesiveness occur, the precise mechanism for them are unknown. Two possibilities are evident: (1) Bonds arise *de novo* in the course of cell differentiation which are responsible for selective adhesion, while pre-existing bonds are retained and are responsible for indiscriminate adhesion. In normal segregation the latter ultimately yield to the increasing tendency for cell movement, and segregation occurs. In experimental cell mixtures, through re-association of differentiated cells, these indiscriminate bonds are once more called into play and yield again. (2) The early bond type (indiscriminate) becomes *modified* according to cell type to the extent that the degree of adhesion varies with the cell type involved. The modification is however not so extreme that indiscriminate adhesions can no longer occur. Again, in conjunction with different degrees of adhesion, changes in tendency for cell movement lead to normal segregation (stages I–IV), while in experimental re-association the condition of less effective cell union is re-introduced (stage III) and segregation occurs once more. Either of these mechanisms may be responsible for the changes of adhesion observed.

At present, it would be futile to speculate further upon the possible subcellular factors that are engaged in cellular adhesiveness. It should be pointed out however that this principle is of universal significance in morphogenesis, and that, in connection with directed cell movements, it is deserving of more attention than it has received.

SUMMARY

1. With the aim of elucidating the factors which are engaged in the morphogenetic movements and tissue segrega-

tions of the amphibian embryo, tissues of known prospective fate from embryos between gastrula and late neurula stages were excised and united with each other in different combinations. The excised tissues were either simply combined in various permutations or they were at first disaggregated into single cells by the addition of alkali, whereupon the cells of one primordium were intermingled with those of another primordium. When returned to optimal pH conditions, the cells united to form a compact body.

2. The mixed-up individual cells perform, according to their cell type, the same kinds of directed movements as do the corresponding tissue fragments. These movements are essentially of two kinds: (1) inward movements, as manifested by the cells from the neural plate, or the mesoderm, which have been combined with either epidermal or endodermal cells; (2) outward movements and peripheral spreading, as manifested by epidermal and, to a lesser extent, by endodermal cells or tissue fragments under the same conditions. It may be concluded that the normal pattern of gastrulation-neurulation movements is primarily due to different migration tendencies inherent in each cell type; that the contractile surface film of the embryo ("coat") merely coordinates but does not — as suggested by Lewis — cause or direct either invagination or epiboly; that the inward movement of single cells or cell sheets does not result from localized "growth" (His), from localized water uptake (Spek, Glaser) or from a pushing effect of the surrounding ectoderm (Giersberg). The migratory cells merely require a rather unspecific cellular substratum to move away from the periphery. The outward movement of epidermal cells in a composite explant can be likewise ascribed to cell-inherent migratory tendencies.
3. The tendency of neurogenic cells to move inward is not confined to material from the neurula stage, but is still present in the explanted and properly combined pieces of brain from larvae. This tendency is absent in the prospective medullary plate prior to induction. Both endodermal or ectodermal cells

represent an adequate embedding substratum for the invading neurogenic material.

4. A neurocoel can be formed (1) as a result of the infolding of any portion of the coated medullary plate, or (2) through secondary cavitation of a compact mass formed by individually immigrated neurogenic cells.

5. In consequence of directed movements, the different cell types in a composite aggregate are sorted out into distinct homogeneous layers, the stratification of which corresponds to the normal germ layer arrangement. The tissue segregation becomes complete because of the emergence of a selectivity of cell adhesion: homologous cells when they meet remain permanently united to form functional tissues, whereas a cleft develops between certain non-homologous tissues (e.g., between neural or endodermal tissues and adjacent epidermis). As in normal embryogenesis, the mesodermal elements take up an intermediate position connecting the inner with the outer epithelia.

6. Regional organ characteristics such as may develop in combinations of non-dissociated fragments hardly ever occur although the reconstituted part-embryos display rather typical cyto- and histodifferentiation. One may conclude that the districts of medullary plate and axial mesoderm are to a certain extent locally determined and that the disaggregated and intermingled cells of these primordia are incapable of reestablishing these patterns. Corresponding experiments on gastrula material yielded more typical neural and mesodermal structures reflecting a higher degree of regulative capacity in these younger stages.

7. It would appear that the principles of cell-specific movements, selective adhesions and cavitation, as manifested in the present experiments, help to elucidate the factors instrumental in the normal formation of germ layers and their further segregation. An interpretation of these phenomena in cytological and physicochemical terms is attempted.

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